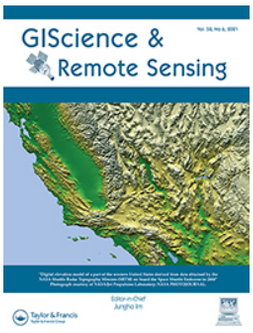

GIScience & Remote Sensing

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/tgrs20

Analyzing spatial variability of drivers of coastal wetland loss in the northern Gulf of Mexico using Bayesian multi-level models

Tyler Hardy, Wei Wu & Mark S. Peterson

To cite this article: Tyler Hardy, Wei Wu & Mark S. Peterson (2021) Analyzing spatial variability of drivers of coastal wetland loss in the northern Gulf of Mexico using Bayesian multi-level models, GIScience & Remote Sensing, 58:6, 831-851, DOI: 10.1080/15481603.2021.1943212

To link to this article: https://doi.org/10.1080/15481603.2021.1943212

View supplementary material

Published online: 01 Sep 2021.

Submit your article to this journal

Article views: 1041

View related articles

View Crossmark data

Citing articles: 2 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tgrs20

GISCIENCE & REMOTE SENSING
2021, VOL. 58, NO. 6, 831–851
https://doi.org/10.1080/15481603.2021.1943212

Taylor & Francis
Taylor & Francis Group

Check for updates

Analyzing spatial variability of drivers of coastal wetland loss in the northern Gulf of Mexico using Bayesian multi-level models

Tyler Hardy, Wei Wu and Mark S. Peterson

Division of Coastal Sciences, School of Ocean Science and Engineering, The University of Southern Mississippi, Ocean Springs, United States

ABSTRACT

A modeling framework that can quantify the impact of multiple potential drivers on coastal wetland loss while estimating its uncertainties is needed to better conserve and restore these valuable ecosystems. We developed a Bayesian multi-level modeling framework to predict the areal wetland loss in the northern Gulf of Mexico, driven by relative sea-level rise (RSLR), vegetation productivity, tidal range, wave height, and coastal slope. We further investigated how the impact of these site or local-scale biogeophysical factors on wetland loss differed by watershed unit. The results indicate the importance of accounting for watershed boundaries where the coastal wetlands are located when evaluating their loss. The effects of RSLR, wave height, and tidal range on wetland loss differed by watershed. Inclusion of streamflow and land use/land cover at the watershed scale did not improve model predictions while other relevant watershed-scale covariates, such as sediment availability, are not available. Based on the 95% credible intervals of the posteriors of the model coefficients, there were only five watersheds among the 14 studied where RSLR, wave height, or tidal range, showed positive or negative effects on coastal wetland loss. The results demonstrate the complexity of modeling wetland loss processes as no single dominant driver was identified for most of the watersheds. Furthermore, the need to account for multiple drivers simultaneously to ensure the effectiveness of costly wetland restoration projects clearly appears warranted when evaluating wetland loss processes over a broad spatial scale.

ARTICLE HISTORY
Received 19 January 2021
Accepted 8 June 2021

KEYWORDS
Bayesian modeling; Scales; Relative sea-level rise; Wave height; Tidal range

1. Introduction

Coastal wetlands offer a variety of ecosystem services, including food production, flood protection, storm surge reduction, water quality improvement, blue carbon sequestration, recreational opportunities, and nursery habitats (Barbier 2013, 2016; Costanza et al. 2008; Engle 2011; Interis and Petrolia 2016; Mendelssohn et al. 2012; Petrolia et al. 2014). However, wetlands have been lost or degraded dramatically in the last few decades (Blum and Roberts 2012; Davidson and Janssens 2006; Templet and Meyer-Arendt 1988; Walker et al. 1987), especially in the northern Gulf of Mexico (NGOM) (Bourne 2000). The extensive loss in the NGOM has been attributed to sea-level rise, land subsidence, reduced sediment input, storm surge, land conversion, levee constructions, and oil spills, to name a few hypothesized causes (Dahl and Stedman 2006; Reed 1995; Turner and Cahoon 1987; Couvillion et al. 2017; Entwistle, Mora, and Knight 2018; Lin and Shen 2018). This loss or degradation results in a reduction in vital ecosystem services (Barbier 2013, 2016; Caffey, Wang, and Petrolia 2014; Costanza et al. 2008; Farber, Costanza, and Wilson 2002; de Groot et al. 2012; Interis and Petrolia 2016; Petrolia et al. 2014). To maintain coastal wetlands in the long term, the surface elevation must be retained through vertical accretion, and lateral erosion must be compensated by sediment supply (e.g., Ganju et al. 2017; Morris et al. 2002). In addition, coastal wetlands may migrate landward if there are no slope or infrastructure barriers with suitable habitats available to compensate for their loss mainly at the water fringe (Enwright, Griffith, and Osland 2016; Borchert et al. 2018).

River sediments delivered into the estuaries provide the main mechanism in the formation and maintenance of coastal wetlands. Studies show salt marshes in marine dominated estuaries with limited sediment input are more vulnerable to relative sea level rise (RSLR) compared to those in riverine dominated estuaries (Alizad et al. 2018; Wu et al. 2020). Vegetation provides another main mechanism that

CONTACT Wei Wu wei.wu@usm.edu
Supplemental data for this article can be accessed here.
© 2021 Informa UK Limited, trading as Taylor & Francis Group

maintains coastal wetlands as it promotes vertical accretion of marsh platforms by trapping water column mineral sediments and contributes organic matter through primary production (e.g., Groffman et al. 2006; Morris et al. 2002; Mudd, Howell, and Morris 2009; Mudd et al. 2013; Nyman et al. 2006). Modeling the dynamics of coastal wetlands across a large spatial scale thus requires knowledge of vegetation characteristics that can be based on remote sensing vegetation indices. For example, the normalized difference vegetation index (NDVI) is commonly used to infer vegetation growth and primary productivity in terrestrial and aquatic ecosystems as it is correlated to productivity and green biomass estimates (Couvillion and Beck 2013; Payero, Neale, and Wright 2004; Wen, Yang, and Saintilan 2012; Hardisky, Gross Gross, and Klemas 1986).

In addition to upland inputs and *in-situ* vegetation productivity, coastal slope, marine related tidal range and wave height affect how coastal wetlands respond to RSLR. For example, flatter terrain, smaller tidal ranges, and larger wave heights all contribute to increased loss of coastal wetlands for a given RSLR (Kirwan and Guntenspergen 2010; Passeri et al. 2015; Karimpour, Chen, and Twilley 2016). These factors need to be accounted for simultaneously in assessing coastal wetland loss. The effects of all of these factors on coastal wetlands, however, likely vary spatially and interact with upland inputs. An effective way to address this is to integrate these factors into a modeling framework that allows inferences and predictions of their contributions to coastal wetland loss spatially.

A variety of models, ranging from simple to complex, have been developed to predict the impact of RSLR on coastal wetland change involving hydrodynamic, geomorphological, and ecological processes (Fagherazzi et al. 2012). While some of these models focus on vertical dynamics of salt marshes (Morris et al. 2002; Marani et al. 2007; Kirwan and Brad Murray 2007; Wu, Biber, and Bethel 2017; Rogers, Saintilan, and Copeland 2012; Stralberg et al. 2011; Schile et al. 2014; Mudd, Howell, and Morris 2009), others focus on lateral erosion due to waves (Mariotti and Fagherazzi 2013; Marani et al. 2011). However, process-based models generally require detailed data inputs and are thus difficult to implement at broad spatial scales as many estuarine systems lack a full data set to apply these models. Simpler models account for the main processes of wetland dynamics, likely in a statistical model or based on the spatial relations of wetland types (Richard et al. 1989; Wu et al. 2015; Ross et al. 2009; LaFever et al. 2007). They are easily applied at a broader spatial scale as they require fewer data inputs, and provide insight for future research approaches on the main loss mechanisms. However, they do not include interactions and feedbacks between geomorphological and ecological processes and therefore could overestimate landscape change (Kirwan and Guntenspergen 2009).

Despite the existence of these models, it is common to relate wetland loss directly to RSLR alone. However, this is problematic due to the inherent complexity and variability of the ecosystem processes that involve multi-scale data and associated processes, and a variety of biogeophysical factors in addition to RSLR (Passeri et al. 2015). Even in the absence of RSLR, mature marshes are lost due to wave-induced edge erosion (Fagherazzi et al. 2013; Mariotti and Fagherazzi 2013) and lack of sufficient sediment supply to compensate for the erosion. As the sediment supply from uplands operates at the watershed scale, a multi-scale geospatial analysis covering the Atlantic and Gulf coasts of the USA indicates that it is important to understand how local processes (specific estuarine sites) operate within heterogeneous broader scales (watersheds) in order to forecast wetland distribution (Braswell and Heffernan 2019). An effective way forward is to assimilate data at multiple spatial scales to understand coastal wetland dynamics and more broadly ecosystem dynamics in general (Peters et al. 2008; Qian et al. 2010).

Here we developed multi-level models in a Bayesian framework to study the spatially-explicit wetland loss driven by RSLR, vegetation productivity (approximated by NDVI), tidal range, wave height, and coastal slope – all interacting with watersheds, meaning we assume the drivers' effects on wetland loss differed by watershed. No comprehensive study has quantified the role of these factors and their interactions with watersheds on coastal wetland loss at the scale of the entire NGOM with consideration of the uncertainties of potential impacts. We attempted to quantify which environmental factors affected coastal wetland loss from 1996 to 2005 and addressed whether the environmental factors had different effects on wetland loss in different watersheds. Given the extensive spatial data sets used and limited

watershed-scale data availability, we consider our research exploratory in nature. However, based on the existing literature on potential drivers of wetland loss, we hypothesized that 1) lower RSLR, higher NDVI, higher tidal range, lower wave height, and a steeper coastal slope, separately or in combination led to lower wetland loss in the NGOM, and 2) the effects of these biogeophysical factors' on wetland loss differed by watershed.

2. Materials and methods

We implemented multi-level models in a Bayesian framework to determine how site-scale vegetation productivity and geophysical variables including RSLR, tidal range, wave height, and coastal slope predictably estimated the loss of coastal wetlands in 14 different watersheds in the NGOM from 1996 to 2005 (before Hurricane Katrina) (Figure 1). We constructed models using all possible combinations of geophysical variables and NVDI as covariates and examined whether we need to account for watershed boundaries when examining the impact of the covariates on coastal wetland loss. This process created 484 models. We then compared the suite of models based on posterior predictive loss (PPL) and deviance information criterion (DIC) to identify the best model(s) and the important biogeophysical variables; the lower the PPL or DIC, the better the model prediction (*sensu* Hooten and Hobbs 2015). Finally, we evaluated whether the inclusion of streamflow and land use/land cover at the watershed scales improved model predictions. Due to the lack of knowledge of multi-scale consideration in modeling coastal wetland loss on the scale we attempted, an exploratory model comparison approach was necessary. The model comparison integrated variable selection and involved comparing models with different combinations of potential site-scale drivers as well as comparing models in which the effect of potential drivers on coastal wetland varied by watersheds vs. not.

2.1. Study area

Our study area is coastal wetlands in the entire NGOM ranging from the western coastline of Florida in the east to the Texas coastline in the west. The majority of the NGOM is humid warm temperate/subtropical; although the southern tip of Florida, including the Florida Keys, has a tropical climate. Mangroves dominate along the southern coast of Florida (Mendelssohn et al. 2017) with *Rhizophora mangle* (red mangrove) dominating shorelines and creek banks, while monospecific stands of *Avicennia*

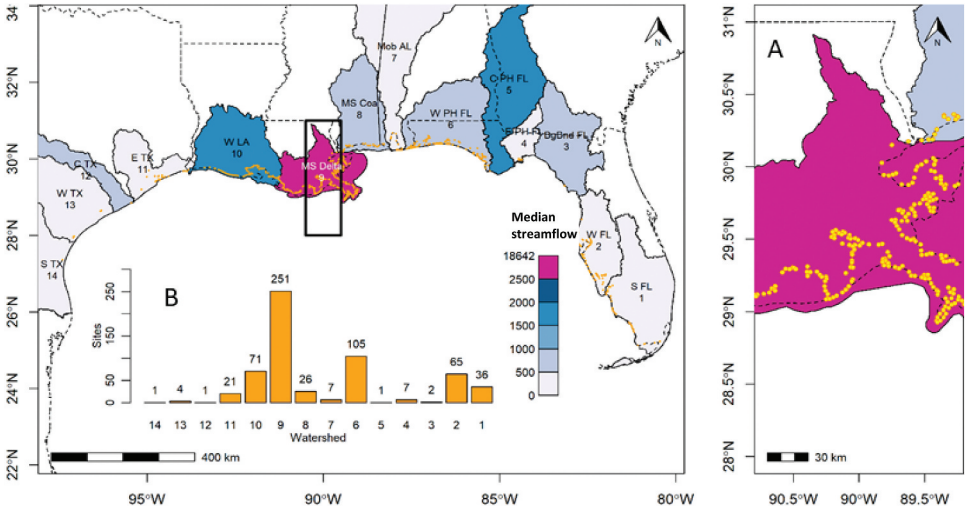

Figure 1. The study area with two spatial scales: watersheds using Hydrological Unit Codes 4 (HUC4) watershed boundaries shown in blue to purple, and sites of wetland loss shown as yellow dots (see enlarged right panel – A). Darker colors of blue or purple indicate higher 10-year summer median of streamflow discharge (ft^3/sec) derived from the USGS streamflow data. Subplot on the bottom (B) shows the number of sites within each watershed. Watershed numbers on the map correspond to the watershed numbers on the subplot. Official HUC4 designations are: 1) Southern Florida (S FL), 2) Peace-Tampa Bay (W FL), 3) Suwannee (Big Bend FL), 4) Ochlockonee (E PH FL), 5) Apalachicola (C PH FL), 6) Choctawhatchee-Escambia (W PH FL), 7) Mobile-Tombigbee(Mob AL), 8) Pascagoula (MS Coast), 9) Mississippi Delta (MS Delta), 10) Louisiana Coastal (W LA), 11) Galveston Bay-San Jacinto (E TX), 12) Lower Colorado-San Bernard Coastal (C TX), 13) Central Texas Coastal (W TX), and 14) Nueces-Southwestern Texas Coastal (S TX) from east to west.

germinans (black mangrove) or mixed stands of both species and/or *Laguncularia racemosa* (white mangrove) are in the landward areas. North of Tampa Bay, salt marsh gradually replaces mangrove (Osland et al. 2013) but due to its cold-tolerant trait (Markley, McMillan, and Jr Thompson 1982), only *A. germinans* is scattered along the northernmost region of the NGOM, including the Louisiana and Texas coasts (Mendelssohn et al. 2017).

The NGOM region contains the largest areas of salt marshes in North America with 55% of United States' total (Mendelssohn et al. 2017). Salt marshes in western Florida, Alabama, and Mississippi are primarily irregularly flooded marshes dominated by *Juncus roemerianus* (black needlerush) with *Spartina alterniflora* (smooth cordgrass) at the seaward of the *Juncus* zone. *Spartina patens* (salt meadow cordgrass) and *Distichlis spicata* (salt grass) are at the higher elevations landward of the *Juncus* zone. Other common salt marsh species include *Spartina cynosuroide*s (big cordgrass), *Salicornia* sp., *Schoenoplectus americanus* (chairmaker's bulrush), and *Schoenoplectus robustus* (sturdy bulrush). *Spartina alterniflora, A. germinans*, and *J. roemerianus* dominate the deltaic marshes of Louisiana with subdominant species of *S. patens* and *D. spicata*. In the higher elevation of the Chenier Plain of southwestern Louisiana and southeastern Texas, the high marsh species, *S. patens* and *D. spicata* dominate brackish marshes. Salt marshes along the central Texas (Galveston Bay and adjacent bays) comprise mainly *S. alterniflora*, then *Batis maritima* (Saltwort), *Salicornia* spp., and *J. roemerianus*. The larger areas of brackish marshes at the higher elevations are dominated by *D. spicata, Salicornia bigelovii* (dwarf saltwort), *Sarcocornia perennis* (chickenclaws), *Monanthochloe littoralis* (shoregrass), and *B. maritima* with occurrence of *S. patens* and *Spartina spartinae* (gulf cordgrass). The marshes in the southern Texas transition to succulent-dominated hypersaline marshes, including *B. maritima, Borrichia frutescens* (sea ox-eye), *Suaeda martima* (herbaceous seepweed), and *Sesuvium portulacastrum* (sea purslane).

2.2. Data

We used five main data sources to develop the models. First, the USGS Hydrological Unit Codes 4 (HUC4) watershed boundaries (https://www.usgs.gov/core-science-systems/ngp/national-hydrography/watershed-boundary-dataset?qt-science_support_page_related_con=4#qt-science_support_page_related_con) to represent the pooled upland inputs (hydrology, sediments, and nutrients). It was used as a random factor to represent one broad spatial scale in the multi-level models (Figure 1). Second, the USGS streamflow data (ft^3/sec) (https://waterdata.usgs.gov/nwis/sw), the only hydro-related data available at all the watersheds in the NGOM, used to represent watershed characteristics and the input to estuaries from uplands. Third, the USGS THK99 dataset (https://pubs.usgs.gov/dds/dds68/htmldocs/data.htm) for the geophysical covariates in the models including long-term average of RSLR rate, tidal range, mean wave height, and coastal slope (Figure 2). Fourth, Landsat-5 satellite images (https://earthexplorer.usgs.gov/) to calculate the normalized difference vegetation index (NDVI) to indicate vegetation productivity. Fifth, the NOAA Coastal Change Analysis Program (CCAP) data to 1) (https://coast.noaa.gov/digitalcoast/data/ccapregional.html) derive areal coastal wetland loss as the response variable in the models, 2) calculate percent developed areas including medium-high developed areas and cultivated lands, and 3) calculate percent forest areas including evergreen, deciduous, and mixed forests averaged for the years of 1996 and 2005 before Hurricane Katrina. Percent developed and forest areas represent watershed-scale anthropogenic influence.

THK99 is a spatially-referenced dataset consisting of long-term (50–100 years) averages of RSLR rate, tidal range, wave height, and coastal slope at ~5 km intervals along the outermost shorelines of the contiguous United States and Alaska. The purpose of the data is to evaluate coastal vulnerability to relative RSLR (https://pubs.usgs.gov/dds/dds68/data/gulf/gulf.htm) and it has been applied to analyze coastal shoreline change (Thieler and and Hammar-Klose 1999; Gutierrez, Plant, and Thieler 2011a). The same drivers affecting shoreline change likely affect coastal wetland erosion. Furthermore, the shorelines the THK99 provides overlap with the boundaries of coastal wetlands, and therefore the data is appropriate for the analysis of vulnerability of coastal wetlands to RSLR. One limitation is that the wave data represents open estuarine conditions and may not accurately represent the wave height at coastal wetlands.

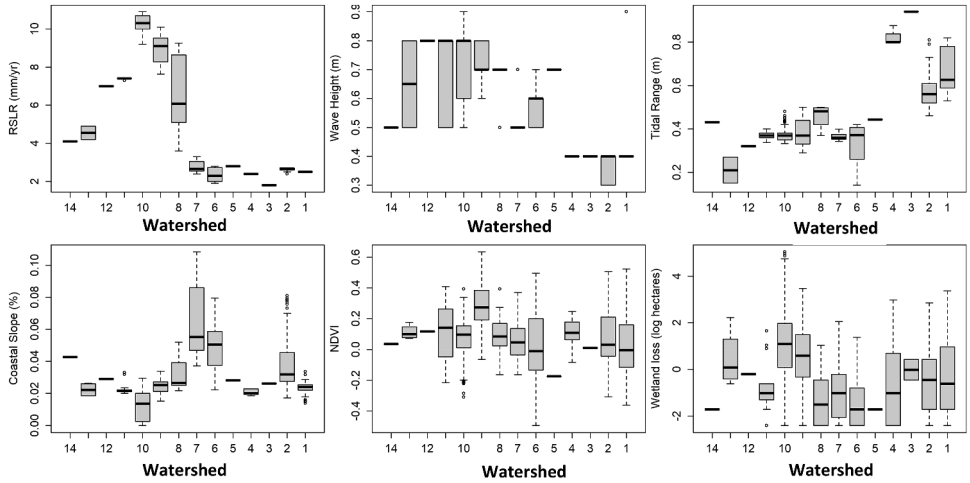

Figure 2. Boxplots of covariates and response variable for each watershed. The thick horizontal line shows the median, the wide box shows the upper and lower quartiles (25 and 75%), and the dotted "whiskers" show the distribution from the quartiles to the 1.5 interquartile range. RSLR denotes relative sea-level rise, and NDVI denotes normalized difference vegetation index from Landsat-5 imagery. Note: Landsat-5 images have seven spectral bands: band 1 = blue; band 2 = green; band 3 = red; band 4 = near-infrared (NIR); band 5 = shortwave infrared (SWIR 1); band 6 = thermal; band 7 = SWIR 2 with a revisit time of 16 days and a spatial resolution of 30 meters (m) except the thermal band with a spatial resolution of 120 m.

However, this is the most consistent and best available wave data at the required region-specific scale and it has been used to classify vulnerability of coastal wetlands (Gutierrez, Plant, and Thieler 2011b).

We generated a point centroid for each 5-km polyline segment from the NGOM mainland shorelines (excluding barrier island shorelines) in the THK99 (n =1,184 segments), and created a circular buffer with a radius of 2.5 km around each centroid. We chose 2.5 km because it was half of the average distance between point centroids, which maximized spatial coverage while minimizing overlaps of buffered areas. We then used the buffer areas as the finest scale for modeling, described as "sites" in this paper. We extracted the geophysical properties at the centroid of each buffer area to represent the properties for that site area. If the buffer areas crossed two watersheds, we did not include them in the final analysis.

Watershed boundary represents a broader spatial scale than the site scale as each watershed contains multiple sites. We used 14 HUC4 watershed boundaries/hydrological regimes from the National Watershed Boundary Dataset (Figure 1) as a master variable to represent the pooled upland inputs, including freshwater, sediment, and nutrients, etc. We assumed the impacts of biogeophyscial variables on coastal wetland loss were more similar within the same watersheds than in different watersheds as the wetland sites in the same watersheds received similar upland inputs. We used the HUC4 to ensure there are enough sites within each watershed in order to obtain robust results on the impact of local drivers on coastal wetlands. We further evaluated whether inclusion of streamflow and land use/land cover at the watershed scales improved model predictions or not if the impact of local drivers differed by watershed.

The biological covariate used was NDVI, which is the normalized difference between the reflectance in the near infrared band and the red band. We downloaded the Landsat-5 images and calculated NDVI values using the Google Earth Engine (GEE) platform. The temporal coverage of the NDVI data was June through August of 1996, the season for peak vegetation productivity in the NGOM (Mo, Kearney, and Turner 2019; Mo, Kearney, and Momen 2017). We created a cloud-free image collection in the GEE for the entire temporal and spatial coverage and used the median NDVI of each 30 m pixel from June to August to generate the final composite image for NDVI. We extracted the NDVI values for only coastal wetland areas using GEE and then calculated the average of NDVI of the coastal wetlands for each study site. Larger NDVI values generally reflect higher aboveground green biomass (Couvillion and Beck 2013; Payero, Neale, and Wright 2004; Wen, Yang, and Saintilan 2012; Hardisky, Gross Gross, and Klemas

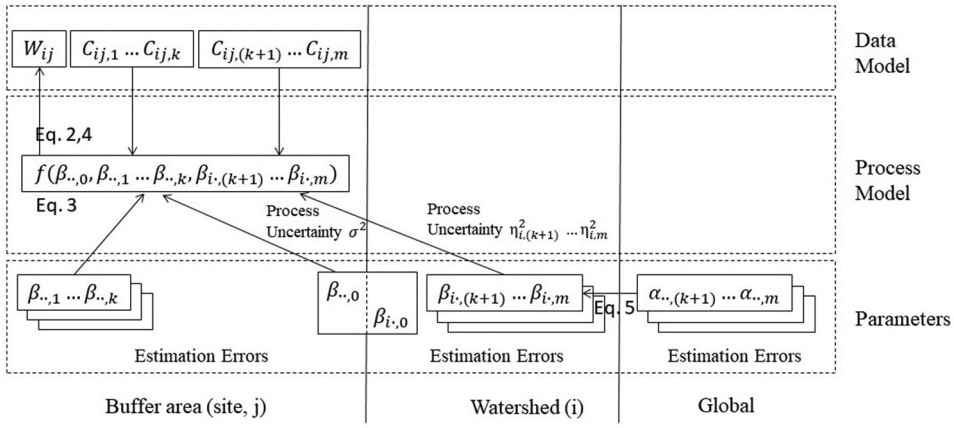

Figure 3. Structure of the Bayesian multi-level model used to identify the biogeophysical variables that affected wetland loss. Wetland loss was a function of geophysical properties and a vegetation index, the effects of which may $(covariates\ k+1 \dots m)$ or may not $(covariates\ 1 \dots k)$ vary in different watersheds. For the covariates that we assumed to affect wetland loss by watershed, we sampled their associated parameters for each watershed from the parameters at the scale of the entire study area (global scale). $m =$ *total number of covariates*, $k =$ *number of covariates*, the effects of which on wetland loss did not vary in different watersheds. $\beta s =$ *intercept and parameters* associated with normalized covariates (C), $as =$ *the hyper − parameters* from which βs were sampled, $W =$ *logarithm of areal wetland loss*, $i =$ *watersheds*, and j = sites. Refer to the main text for formulas of different equations identified in this figure.

1986). We assumed healthy vegetation in 1996 helped maintain coastal wetlands for the next decade based on the observation that vegetation cover in prior year was the best single predictor of subsequent wetland loss in coastal Louisiana (Schoolmaster et al. 2018).

We derived the response variable, indicating areal wetland loss (ha) between 1996 and 2005, from NOAA's CCAP data for the years of 1996 and 2005 (before Hurricane Katrina). CCAP land use/land cover data has a spatial resolution of 30 m, an appropriate resolution for a broad scale like the entire NGOM. It is likely this data may not capture coastal wetland loss finer than this resolution. We calculated areal wetland loss as the sum of cells, converted from any estuarine or palustrine wetlands to open water, at each site multiplied by the area of each cell. This estimate excluded the buffer areas without wetland loss that mainly represent areas with restoration activities or very limited coastal wetland areas. In total, we had 598 sites with wetland loss (Figure 1), and Figure 2 illustrates the five biogeophysical covariates and response variable by watershed.

2.3. Multi-level modeling in a Bayesian framework

Hierarchical Bayesian models allow the assimilation of data at multiple spatial scales through multi-level modeling and can quantify uncertainties (Clark 2005; Wu et al. 2012). They accommodate complexity by decomposing high-dimensional relationships into levels of conditional distribution within a consistent framework (Clark 2001; Wu, Clark, and Vose 2010). The framework includes a data level, a process model level, and a parameter level. In this framework, numerous latent variables and parameters that describe complex relationships are quantified in the form of posterior probability distributions that make uncertainty readily derived using credible intervals (Clark and Gelfand 2006). The Bayesian method usually applies Markov Chain Monte Carlo (MCMC) simulations to sample posterior distributions based on the product of likelihood and prior probability (Gilks, Richardson, and Spiegelhalter 1996).

Due to the multi-scale data (site and watershed scales) involved in analyzing coastal wetland loss, we used a multi-level modeling approach that is powerful when analyzing data based on nested designs (Bolker et al. 2009; Gelman and Hill 2007; Wu et al. 2018). We focused on studying the cross-scale interaction by examining how the effects of site-scale covariates on coastal wetland loss differed by watershed. The Bayesian inference accounts for uncertainties from data, processes, and parameters (Figure 3). Though some watersheds had only one site or a small numbers of sites, the variances could still be estimated due to the partial pooling with the multi-level models (Kéry 2010; Technow and Radu Totir 2015), but generally large. Partial pooling allows estimation of

watershed-specific effects and the "borrowing" of information across watersheds. To represent the function of wetland loss (i.e., logarithm of wetland loss) at the watershed *i* and site *j* (W_{ij}), let $W_{ij} \cdot \mu$ represent the mean of W_{ij}, and σ^2 represent the variance of wetland loss across sites. W_{ij} was modeled by assuming it was distributed as $(\sim)$a normal distribution (N) (Equation 1):

$$W_{ij} \sim N(W_{ij.}\mu, \sigma^2). \tag{1}$$

We modeled the mean wetland loss $(W_{ij.}\mu)$ using a linear function of the covariates C_1-C_m, with *k* covariates having the same effect on wetland loss across the entire study area, and the remainder of the covariates $(m-k)$ having different effects on wetland loss in different watersheds. The specific functions representing the relationships between the potential drivers and coastal wetland loss are likely to vary by watershed, so we believe the linear representation of these relationships is necessary and appropriate at the scales we focused on for this study (*sensu* Braswell and Heffernan 2019). The intercept may vary $(\beta_{i.,0})$ or not vary $(\beta_{..,0})$ across watersheds. Assuming the intercept did not vary across the watersheds, we described the linear function using Equation 2:

$$\begin{aligned} W_{ij}.\mu &= f\left(\beta_{..,0}, \beta_{..,1} \ldots \beta_{..,k}, \beta_{i.,(k+1)} \ldots \beta_{i.,m}\right) \\ &= \beta_{..,0} + \sum_{l=1}^{k} \beta_{..,l} C_{ij,l} + \sum_{l=k+1}^{m} \beta_{i.,l} C_{ij,l} \end{aligned} \tag{2}$$

where βs represents the parameters for the linear model. If the intercept varied across watersheds, $\beta_{..,0}$ should be replaced with $\beta_{i.,0}$ in Equation 2. $\beta_{..,l}$ represents the coefficient for $C_{ij,l}$, for which the effects on wetland loss did not vary across the watersheds ($1 \le l \le k$). $\beta_{i.,l}$ represents the coefficient for $C_{ij,l}$, for which the effects on wetland loss varied across the watersheds ($k + 1 \le l \le m$). Therefore, we modeled wetland loss (**W**) for the R watersheds and S_i sites in each watershed (i) was modeled as in Equation 3:

$$\begin{aligned} &p\left(\mathbf{W} | \beta_{..,0}, \beta_{..,1} \ldots \beta_{..,k}, \beta_{i\cdot,(k+1)} \ldots \beta_{i\cdot,m}\right) \\ &\propto \prod_{i=1}^{R} \prod_{j=1}^{S_i} N\left(W_{ij} | f\left(\beta_{..,0}, \beta_{..,1} \ldots \beta_{..,k}, \beta_{i.,(k+1)} \ldots \beta_{i.,m}\right), \sigma^2\right). \end{aligned} \tag{3}$$

We sampled the coefficients that varied across the watersheds $(\beta_{i.,(k+1)} \ldots \beta_{i.,m})$ from the parameters at the coarser gulf-wide scale (or global scale in Figure 3) using normal distributions with means of αs and variances among watersheds of η^2 (Equation 4):

$$\begin{aligned} &\beta_{i.,(k+1)} \sim N\left(\alpha_{..,(k+1)}, \eta^2_{(k+1)}\right) \\ &\beta_{i.,(k+2)} \sim N\left(\alpha_{..,(k+2)}, \eta^2_{(k+2)}\right) \\ &\ldots \\ &\beta_{i.,m} \sim N\left(\alpha_{..,m}, \eta^2_{m}\right). \end{aligned} \tag{4}$$

To complete the Bayesian model, we defined prior distributions for unknown parameters $(\alpha s, \beta s - \beta_{..,1} \text{ to } \beta_{..,k}, \sigma^2, \text{ and } \eta^2 s)$. We used conjugate prior for computation efficiency (Calder et al. 2003) and therefore the priors and posteriors had the same probability distribution forms. The priors for *αs* and *βs* were normally distributed, and the priors for σ^2 and $\eta^2 s$ followed the inverse gamma distributions. The prior distributions were flat and only weakly influenced the posteriors, which reflected the lack of knowledge of these parameters (Lambert et al. 2005) (see supplementary material for the specification of the priors).

By combining the parameters (priors), processes, and data models, we derived the joint posterior distribution of unknowns (Clark 2005) in Equation 5:

$$p\left(\begin{matrix} \beta_{..,0}, \beta_{..,1} \ldots \beta_{..,k}, \beta_{i.,(k+1)} \ldots \beta_{i.,m} \\ \alpha_{..,(k+1)} \ldots \alpha_{..,m}, \\ \sigma^2, \eta^2_{(k+1)}, \ldots, \eta^2_{m} \end{matrix} \middle| \boldsymbol{W}, \boldsymbol{C_1} \ldots \boldsymbol{C_m}\right) \tag{5}$$

$$\begin{aligned} &\propto p\left(\boldsymbol{W} | \boldsymbol{f}\left(\beta_{..,\boldsymbol{0}}, \beta_{..,\boldsymbol{1}}, \ldots, \beta_{..,\boldsymbol{k}}, \beta_{\boldsymbol{i}..,(\boldsymbol{k+1})}, \ldots, \beta_{\boldsymbol{i}..,\boldsymbol{m}}\right), \sigma^{\boldsymbol{2}}\right) \\ &\times \left(p\left(\beta_{i.,(k+1)} | \alpha_{..,(k+1)}, \eta^2_{(k+1)}\right) \times \ldots \times p\left(\beta_{i.,m} | \alpha_{..,m}, \eta^2_{m}\right)\right) \\ &\left.\begin{aligned} &\times p(\beta | ..,0) \times \left(p\left(\beta_{..,1}\right) \times \ldots \times p\left(\beta_{..,k}\right)\right) \times p\left(\sigma^2\right) \\ &\times \left(p\left(\alpha_{..,(k+1)}\right) \times \ldots \times p\left(\alpha_{..,m}\right)\right) \\ &\times \left(p\left(\eta^2_{(k+1)}\right) \times \ldots \times p\left(\eta^2_{m}\right)\right) \end{aligned}\right\} \text{Priors} \end{aligned}$$

The model in Equation 5 was parameterized using MCMC simulations (Gelfand and Smith 1990) in JAGS through the package "rjags" (Plummer 2016) in CRAN R (R Core Team 2015) (see supplementary material for the JAGS code). Prior to implementation of the multilevel models, we performed data transformations to

improve the model computation efficiency. The covariates were normalized using classic standard score normalization (Zar 2010). Normalization allowed comparison of the covariates' impact on wetland loss by examining the magnitudes of their coefficients directly because they were transformed to the same ranges (Hooten and Hobbs 2015). Furthermore, the response variable was log transformed to fit a normal distribution and account for nonlinearity and non-additive effect of the biogeophysical variables. We checked for multicollinearity using variance inflation factor (VIF). We only removed a covariate from further consideration if the VIF for it was greater than five (Zuur et al. 2009). None of the covariates showed VIF >5, so we did not remove any of the covariates.

We simulated three MCMC chains using three different sets of initial values (based on vague priors) for the parameters. The three chains were used to determine the number of iterations of MCMC before they converged for the model (20,000 iterations) (Cowles and Carlin 1996). We then discarded the pre-convergence, burn-in iterations, and ran the chains additional 300,000 iterations thinning every 10 steps to reduce within-chain autocorrelation. The generated posteriors helped identify the covariates that had effects on wetland loss different from zero, evaluate relative magnitude of the effect of each covariate on wetland loss, and determine whether a covariate had different effects on wetland loss in different watersheds.

2.4. Model comparison

We compared the 484 models, based on all combinations of covariates and whether they interacted with watersheds or not, to identify the best model(s) using both deviance information criterion (DIC, Eq. 6) and posterior predictive loss (PPL, Eq. 7) functions. The lower the DIC or PPL, the better the model predicts (Hooten and Hobbs 2015). Although PPL functions measure the error in model predictions as loss (Gelfand and Ghosh 1998; Ibáñez et al. 2009), DIC generally outperforms PPL in comparing models with large sample size (n >≈ 1000) (Daniels, Chatterjee, and Wang 2012). However, DIC only allows for comparison of models with the same level of structures (Hooten and Hobbs 2015). Furthermore, models are considered comparable if the difference of their DICs is less than 2 (Burnham, Anderson, and Huyvaert 2011; Burnham and Anderson 2004). Therefore, we used DIC to compare models with the same hierarchies (i.e., comparing models considering watersheds' effect but with different covariates) and PPL to compare the models with different structures (i.e., comparing models considering watershed's effect to those without). We identified the best model with the lowest PPL and DIC, while considering models as the best candidate models if their DICs are not larger than the DIC of the best model by more than 2 (Hobbs and Hooten 2016).

$$DIC = -2\log([\boldsymbol{W}|E(\boldsymbol{\theta}|\boldsymbol{W}]) + 2p_D \tag{6}$$

where p_D represents an effective number of parameters, $\boldsymbol{\theta}$ represents the parameters in the model, E represents expected values, and [] represents probability.

$$PPL = \sum_{i=1}^{n} \left(y_i - E\left(W_i^{new}|\boldsymbol{W}\right)\right)^2 + \sum_{i=1}^{n} Var\left(W_i^{new}|\boldsymbol{W}\right) \tag{7}$$

where W_i^{new} represents a new data point (wetland loss) simulated from the model, and Var represents variance.

If the best model(s) showed the impact of the covariates or intercept varied by watershed, we further added mean daily streamflow (ft^3/sec), and/or land use/land cover as covariates at the watershed scale and investigated whether the inclusion of them improved the model predictions. In the models that accounted for these watershed-scale covariates, we modeled the mean of $\beta_{i.,0}\left(\beta_{i.,0}.\mu\right)$as a function of streamflow or streamflow and land use/land cover (Equations 8–10). There was no multicollinarity problem when we integrated streamflow and land use/land cover as the highest correlation was −0.37 between percent forest areas and percent developed areas. We compared these different models to the best model(s) based on DIC only as they had the same hierarchies.

$$\beta_{i.,0}.\mu = k_0 + k_1 SF, \tag{8}$$

$$\beta_{i.,0}.\mu = k_0 + k_1 SF + k_2 DV + k_3 FR, \tag{9}$$

$$\beta_{i.,0} \sim N\left(\beta_{i.,0}.\mu, \eta_0^2\right) \tag{10}$$

where k_0-k_3 represent the parameters, SF represents streamflow, DV represents % developed areas, FR

represents % forest areas at the watershed scales, and η_0^2 represents variance of $\beta_{i..0}$.

We examined the 95% credible intervals (CIs) of the posteriors for the coefficients to determine whether the covariates had effect on wetland loss different from zero (not overlapping 0) with the sign of the CI for a given covariate indicating a positive (CI is positive without overlapping zero) or negative (CI is negative without overlapping 0) effect on wetland loss.

We considered Bayesian p-values (Equation 11) as a method to examine the model fitting in moments and quantiles. Bayesian p-values describe the proportion of times the statistic of interest generated from the model is equal to or more extreme than the statistic from the data; thus, they differ from traditional p-values. The value of 0.5 indicates a good fit of the model to the data while the values close to 0 or 1 indicate a lack of model fit that cannot be attributed to chance.

$$\begin{aligned} P &= \mathrm{p}\left[T\left(y_{rep},\theta\right) \geq T(y,\theta)|y\right] \; or \\ &= \mathrm{p}\left[T\left(y_{rep},\theta\right) \leq T(y,\theta)|y\right] \end{aligned} \tag{11}$$

where P denotes p-value, $T(y,\theta)$ is a test-statistic of interest (i.e., mean, standard deviation, quantiles) calculated from observed data (y), while $T\left(y_{rep},\theta\right)$ is a corresponding statistic from the new "data" $\left(y_{rep}\right)$ generated from the posterior predictive distribution. These estimates allow for the calculation of p-values for various statistics (Meng 1994). Table S1 provides all of the structures, DICs, PPLs, and Bayesian p-values for the 484 models.

3. Results

We documented four important findings. First, the impact of RSLR, tidal range, and wave height on wetland loss differed by watershed. Second, NDVI's impact on areal wetland loss did not vary by watershed and its effect was not different from 0. Third, there were only five watersheds among the 14 studied where some of these biogeophysical factors showed positive or negative effects due to the high variances of their impacts on coastal wetland loss. Finally, with variances of the impact of the biophysical factors accounted for, coastal wetland loss was generally attributed to multiple drivers as it was difficult to identify one single dominant driver.

Among the five best model candidates in which ΔDIC is < 2 (out of the 484 models), RSLR, wave height, and tidal range were consistently included, and their effects on wetland loss varied across watersheds (Table 1). The analysis only selected NDVI twice and its effect on wetland loss did not appear to vary across watersheds. The intercept also varied across the watersheds for the best candidate models, indicating varying base wetland loss in different watersheds without accounting for the effect of the covariates.

The model with the lowest PPL and DIC was model 58 (DIC = 2037.59, PPL = 952.27, Table 1, Table S1). Its Bayesian p-values based on mean, variance, and coefficient of variance were 0.50, 0.57, and 0.28, respectively, indicating a good fit for mean and variance but a moderate fit for coefficient of variance (Table S1). RSLR, wave height, tidal range were included as the covariates in this model and their effects on wetland loss varied across the watersheds (Table 1). Based on the medians of the posteriors, most of the watersheds

Table 1. Results of the best model candidates in which ΔDIC is less than 2 out of the 484 models for areal wetland loss. β_0 denotes intercept, RSLR denotes relative sea level rise, WH denotes wave height, TR denotes tidal range, CS denotes coastal slope, and NDVI denotes normalized different vegetation index. DIC denotes deviance information criterion and PPL denotes posterior predictive loss. Both DIC and PPL were applied in the model selection. The lower the DIC and PPL, the better the model predictions. The model numbers under the significant covariates columns denote watersheds identified in Figure 1 where that particular covariate had an effect different from zero (95% credible intervals do not overlap 0) on coastal wetland loss. An X mark under a covariate represents that covariate was included in the model.

	Covariates effects of which do not vary across watersheds						Covariates effects of which vary across watersheds						Model metrics		Significant covariates					
Model #	RSLR	NDVI	WH	TR	CS	β_0	RSLR	NDVI	WH	TR	CS	β_0	ΔDIC	PPL	RSLR	NDVI	WH	TR	CS	β_0
58							X		X	X		X	0.00	952.27	6,8,10		1,9	1,8,9		8
146							X		X	X	X	X	0.08	953.16	6,8,10		1,9	1,9		9
145					X		X		X	X		X	1.46	968.74	6,8,10		1,9	1,9		8,11
241		X					X		X	X	X	X	1.56	969.78	6,8,10		1,9	1,9		8
161		X					X		X	X		X	1.65	968.85	6,8,10		1,9	1,9		8

Table 2. Summary of the coefficients for the covariates selected in the best areal wetland loss model #58. We grouped rows by watershed (denoted by 1 to 14 within model). We highlighted the covariate with the largest absolute median for each watershed. Sd denotes standard deviation, and 2.5% and 97.5% denote 2.5% quantile and 97.5% quantile. Covariate abbreviations defined in Table 1.

	mean	sd	2.5%	median	97.5%
bRSLR[1]	−0.41	0.48	−1.37	−0.4	0.51
bTR[1]	−0.71	0.29	−1.28	−0.71	−0.16
bWH[1]	−0.76	0.27	−1.3	−0.76	−0.22
bRSLR[2]	1.02	0.7	−0.32	1	2.46
bTR[2]	0.09	0.26	−0.42	0.09	0.6
bWH[2]	−0.53	0.43	−1.39	−0.52	0.28
bRSLR[3]	0.47	0.93	−1.49	0.5	2.24
bTR[3]	0.11	0.42	−0.74	0.11	0.93
bWH[3]	−0.25	0.79	−1.87	−0.23	1.28
bRSLR[4]	0.23	0.93	−1.79	0.29	1.93
bTR[4]	−0.22	0.45	−1.14	−0.21	0.65
bWH[4]	−0.45	0.77	−2.07	−0.41	0.99
bRSLR[5]	0.82	0.73	−0.6	0.81	2.28
bTR[5]	−0.37	0.98	−2.37	−0.36	1.57
bWH[5]	−0.17	0.8	−1.8	−0.16	1.38
bRSLR[6]	0.93	0.25	0.45	0.93	1.42
bTR[6]	−0.02	0.16	−0.33	−0.02	0.3
bWH[6]	0.02	0.31	−0.59	0.02	0.63
bRSLR[7]	0.77	0.53	−0.26	0.76	1.82
bTR[7]	−0.54	0.89	−2.39	−0.51	1.16
bWH[7]	−0.11	0.62	−1.35	−0.11	1.1
bRSLR[8]	1.12	0.53	0.1	1.11	2.19
bTR[8]	−1.32	0.65	−2.7	−1.28	−0.12
bWH[8]	−0.44	0.59	−1.64	−0.42	0.71
bRSLR[9]	1.24	0.39	0.48	1.24	2.02
bTR[9]	0.15	0.32	−0.47	0.14	0.78
bWH[9]	0.1	0.29	−0.47	0.1	0.67
bRSLR[10]	0.5	0.3	−0.11	0.5	1.08
bTR[10]	−1.34	0.36	−2.06	−1.34	−0.64
bWH[10]	0.62	0.1	0.43	0.62	0.81
bRSLR[11]	0.27	0.97	−1.84	0.32	2.03
bTR[11]	0.85	0.8	−0.55	0.8	2.56
bWH[11]	0.07	0.3	−0.52	0.07	0.65
bRSLR[12]	0.64	0.97	−1.33	0.65	2.55
bTR[12]	−0.3	0.87	−2.01	−0.31	1.44
bWH[12]	−0.14	0.7	−1.54	−0.13	1.23
bRSLR[13]	0.5	0.89	−1.33	0.52	2.25
bTR[13]	−0.51	0.46	−1.42	−0.51	0.41
bWH[13]	0.25	0.54	−0.8	0.24	1.35
bRSLR[14]	0.88	0.85	−0.78	0.86	2.62
bTR[14]	−0.37	0.98	−2.36	−0.36	1.58
bWH[14]	0.04	0.74	−1.42	0.03	1.54

had RSLR having the largest median effect on coastal wetland loss (Table 2). With variances accounted for, RSLR positively affected wetland loss in Watersheds 6, 8, and 10 (Choctawhatchee-Escambia also known as western Florida Panhandle, Pascagoula also known as MS coast, and Louisiana Coastal also known as western Louisiana or Chenier Plain, respectively) (Figure 4). RSLR had a marginally positive impact on wetland loss in Watershed 9 (MS Delta). Although RSLR's impact was not all different from zero, the medians for the RSLR's coefficients for all the watersheds were positive, except for Watershed 1 (Southern Florida). Tidal range was negatively correlated with wetland loss in Watershed 1, 8, and 9 (Southern Florida, Pascagoula, and MS Delta, respectively), indicating a high tidal range decreased wetland loss in those watersheds. Wave height was negatively related to wetland loss in Watershed 1 (Southern Florida), but positively affected wetland loss in Watershed 9 (MS Delta). Wave height also exhibited a low posterior variance for the MS Delta watershed, and it showed different effect on coastal wetland loss between the MS Delta watershed and watersheds 1 and 2 (southern and western FL, respectively). The intercepts were negative or marginally negative in Watershed 8 (MS coast) and Watershed 11 (Galveston Bay-San Jacinto also known as Eastern Texas) respectively, showing relatively low base wetland loss in these two watersheds. The larger intercepts (both medians and lower 2.5 percentiles) were in Watershed 1 (southern Florida), 9 (MS delta), and 10 (LA coastal (Chenier Plain)), showing higher base wetland loss there. Model 146's DIC was larger than that of Model 58 by only 0.08. Coastal slope was included in Model 146 and it showed negative effect on coastal wetland loss in LA coastal (Chenier Plain) (#10). The posteriors of the other coefficients were similar to those of Model 58. The high variances of these coefficients led to that there were only five watersheds where some of these biogeophysical factors showed effects different from zero on wetland loss. Therefore, we attributed wetland loss, a complex process, to multiple biogeophysical factors. For example, RSLR, wave height and tidal range all played important roles in affecting wetland loss in Watershed 9 (MS Delta). Both RSLR and tidal range had effects on wetland loss different from zero in Watershed 8 (MS coast) (Figure 4).

Though a smaller number of sites in some watersheds could likely lead to effects of the covariates not different from zero (Dick and Tevaearai 2015), this did not explain the lack of effects of the covariates in the majority of watersheds. The three watersheds in the north-central and north-eastern Gulf of Mexico that showed positive effect of RSLR on wetland loss (Choctawhatchee-Escambia #6, MS coast #8, and LA coastal #10) had both high and low number of sites (n = 26, 71, and 105, respectively in Figure 1), when compared to other watersheds. Additionally, the MS Delta watershed (#9) had 251 sites but the coefficient of RSLR showed only marginal effects. For a particular covariate, we drew the posteriors from different

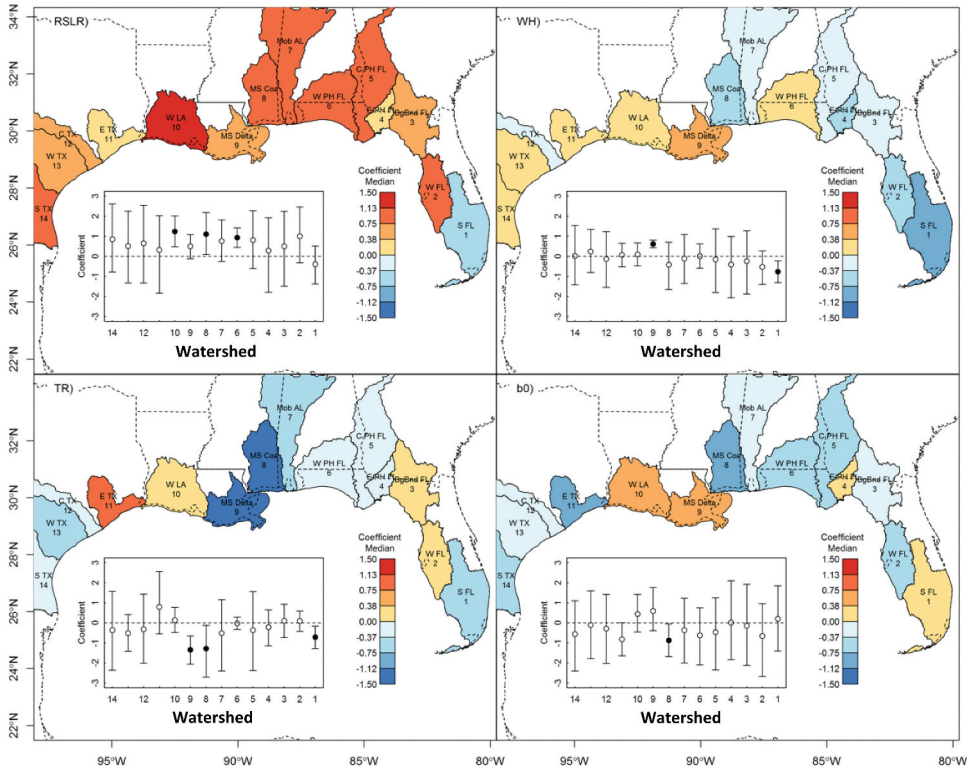

Figure 4. Maps of the medians of the coefficients in the best areal wetland loss model (#58) at the watershed scale (note each watershed has multiple coastal wetland sites). We standardized the covariates so that the coefficients were comparable both between watersheds, and between covariates. In the inset plots, a dot represented the median of the parameter posterior for each watershed, with associated 95% credible interval represented by a line crossing the median. A solid dot indicated a positive or negative effect of the covariate on coastal wetland loss for the given watershed (95% credible interval does not contain 0), and an unfilled dot indicated an effect not different from 0. RSLR denotes relative sea-level rise, WH denotes wave height, TR denotes tidal range, and b0 denotes intercept of the site-scale submodel (Eq. 3).

watersheds from a common distribution, thus their estimates shrank to the common mean. In addition to the effects (different from zero or not), the effect sizes and variances derived from the posteriors also provide important information on the effects of the potential drivers on coastal wetlands (Figure 4).

The best model did not include NDVI; however, NDVI was selected in Model 241 and 161 (DIC = 1.56, 1.65) in the best model candidates. In addition to the covariates included in the best model (Model 58), Model 241 included NDVI as a gulf-wide effect, and coastal slope as a watershed-variant effect. The gulf-wide NDVI parameter was not different from zero, though the median was positive. Compared to the best model (Model 58), the coefficients of RSLR, tidal range, and wave height generally had similar posteriors in Model 241, though the coefficient for RSLR at the Watershed 8 (MS coast) was no longer different from zero (Table 1). In addition, the coefficient for wave height at the Watershed 9 (MS delta) was smaller in magnitude and no longer different from zero in Model 241.

Additionally, the inclusion of streamflow and/or land use/land cover in the models did not improve

Table 3. Comparison of models with streamflow and/or land use/land cover as covariates at the watershed scale for intercept to the best model (Model #58) without consideration of streamflow and/or land use/land cover for intercept. (SF denotes streamflow (ft^3/sec), DV denotes percent developed areas, FR denotes percent forest areas, 2.5% and 97.5% denote 2.5% quantile and 97.5% quantile of the parameters, k_1 denotes the coefficient for streamflow, k_2 denotes the coefficient for percent forest areas, and k_3 denotes the coefficient for percent developed areas).

Model metrics		Parameters			
Model #	ΔDIC	Name	2.5%	Median	97.5%
Model 58	0.00	No covariates at the watershed scale			
SF	−0.75	k_1(SF)	−0.292	0.351	1.020
SF, FR, DV	−1.21	k_1(SF)	−0.289	0.388	1.082
		k_2(FR)	−1.468	−0.590	0.356
		k_3(DV)	−1.181	−0.336	0.517

predictions based on very small changes of DICs (Table 3). Their impact on coastal wetland did not differ from zero (Table 3). This indicates that streamflow and/or land use/land cover did not represent watershed characteristics that differentiate wetland loss across watersheds. Unfortunately, although sediment yield at the watershed scale is more relevant to wetland loss, the data are not available, and should be explored more in future studies.

4. Discussion

We applied multi-level Bayesian modeling to explore and identify how biogeophysical factors affected coastal wetland loss in the entire NGOM by coherently assimilating data at multiple spatial scales. The statistical models applied in our study are well suited for broad-scale analysis due to the low data requirement and spatial data readily available. Few studies exist on linking coastal wetland loss to a variety of important biogeophysical variables at a scale as broad as the entire NGOM. The novelty of the work lies in the two main aspects: 1) we not only identified key local drivers for coastal wetland loss, but also studied wetland land loss in the context of the broader watershed spatial scale; and 2) we quantified the uncertainties related to the impact of potential drivers on coastal wetland loss. Though it is well understood that sea-level rise and waves contribute to wetland loss, a multi-scale consideration in wetland loss is rare. We provided variance information (Figure 4) which is key to understanding coastal wetland dynamics but is largely lacking in the current literature. Therefore, we believe our research improves the general understanding of the controls on coastal wetland loss by addressing these two knowledge gaps, though it remains a challenge to develop adequate predictive models for a process as complex as wetland loss, particularly when important data sets are limiting (e.g., sediment input) and the scale of study is broad.

Multi-scale considerations

In the limited research of coastal wetland dynamics that account for multiple spatial scales, Braswell and Heffernan (2019) focused on wetland distribution at the estuary and watershed scales instead of wetland loss studied here. It did not address or quantify the variance of the impact of the biophysical factors on coastal wetlands either. Their results are, however, useful as they indicated that characteristics of estuaries (depth, shoreline complexity, land use, tidal range, and latitude) and their watersheds (discharge) best predicted wetland extent at broad spatial scales (~>10^2 km^2). In their model, bimodal distribution of wetland extent occurs at finer scales (~10^0 to 10^2 km^2) that correspond to the theoretical scale of wave erosion feedbacks. These results suggest that local feedbacks are the mechanism that controls marsh dynamics, whereas coastal and watershed characteristics determine macroscale wetland distributions through their effects on these local processes.

The interactions between the two spatial scales selected in our final models pinpoint the importance of accounting for watershed boundaries in evaluating coastal wetland loss as upland and marine processes influence coastal wetlands in transitional environments. The coastal sites within the same watershed tend to receive similar upland inputs like decreased freshwater, pollutant discharges, land use/land cover associated changes, and changing climate responses compared to sites in the other watersheds (Hovenga et al. 2016; Alber 2002; Lin and Shen 2018; Huang et al. 2016; Li et al. 2018). Therefore, the watershed boundary in our model coarsely reflects the conditions of upland inputs. The changes in upland inputs affect salinity, nutrients, and temperature dynamics in the systems (Kennedy and Barbier 2016; Elsey-Quirk et al. 2019) and further influence the health of coastal wetlands in the short and long terms. Importantly, upland watershed modifications reduce or eliminate river-borne sediments which are the main contributor of vertical accretion, formation, and maintenance of coastal wetlands in the NGOM or elsewhere (Karegar, Dixon, and Malservisi 2015; Tweel and Turner 2012; Weston 2014; Blum and Roberts 2009). In fact, the fluvial sediment input to the NGOM substantially decreased from 1950 to 2010 due to reduction in agricultural soil erosion (Syvitski et al. 2005), damming of rivers (Vörösmarty et al. 2003), and water-use in the upper watershed (Meade and Moody 2010), which in turn causes extensive loss of coastal wetlands in the NGOM. Because sediment input data is limited and inconsistent across our large study area, including streamflow and land use/land cover as covariates at the watershed scale likely captures some of the

sediment load and nutrient inputs that would represent upland inputs and thus may influence our model results. However, when we accounted for streamflow and land use/land cover in the linear models at the watershed scale, the model predictions did not improve. This is likely due to these two covariates not correlating well with sediment load due to hydrological alterations spatially and temporally on the rivers. The models should also consider nonlinear functions at the watershed scales.

Drivers for coastal wetland loss

The coefficient for RSLR had both a positive median value for each watershed except for southern Florida (#1). RSLR also had the largest median effect on wetland loss among all the covariates in the majority of watersheds (Figure 4). These results are supported by studies that indicated an increase of wetland loss due to increasing sea level (Kirwan and Patrick Megonigal 2013; Linhoss et al. 2015; Morris et al. 2002; Simas, Nunes, and Ferreira 2001; Wu et al. 2015; Wu, Biber, and Bethel 2017). However, large spatial variability existed in the impact of RSLR on areal wetland loss across the NGOM. The north-central and northeastern Gulf of Mexico, including Choctawhatchee-Escambia watershed in western Florida panhandle (#6), Mobile-Tombigbee watershed in Alabama (#7), Pascagoula watershed on MS coast (#8), MS Delta watershed in Louisiana (#9), and LA coastal (or western Louisiana or Chenier Plain) watershed (#10) from east to west, showed highest vulnerability to RSLR based on the posteriors of the coefficient. The patterns we identified are supported by research that indicated high RSLR caused increases in coastal wetland inundation (Alizad et al. 2016), non-linear expansion of tidal creeks (Hagen et al. 2012), and coastal wetland loss (Geselbracht et al. 2015; Glick et al. 2013; Kidwell et al. 2017; Martin and White 2000; Passeri et al. 2016; Walsh 2007).

We expected the RSLR coefficients to be positive and largest for both the MS Delta watershed (#9) and LA coastal (or Chenier Plain) watershed (#10) due to the extensive literature on the subject. For example, high rates of wetland loss was due to high RSLR from sediment compaction and subsidence in Louisiana (Georgiou, FitzGerald, and Stone 2005; Karegar, Dixon, and Malservisi 2015; Mallman and Zolback 2007; Törnqvist et al. 2008; Yuill, Lavoie, and Reed 2009; Couvillion et al. 2017; Bourne 2000). In fact, the medians of the coefficients for RSLR in both Louisiana's watersheds were positive and RSLR showed positive and marginally positive effects in the LA coastal (or Chenier Plain) watershed (#10) and MS Delta (#9) watersheds, respectively. The RSLR's median effect on wetland loss in the LA coastal (or Chenier Plain) watershed was larger than in all other watersheds. This result was consistent with the previous research that showed the LA coastal (or Chenier Plain) watershed was more vulnerable to RSLR compared to the MS Delta due to differences in geomorphological features and processes like proximity to riverine sediment inputs, connectivity to the Gulf of Mexico, impact of Chenier ridges and impoundments (Jankowski et al. 2017).

The coefficient for wave height (Figure 4) was positive only in the MS Delta watershed (#9), indicating that wave height had a positive effect on wetland loss that was comparable to that of RSLR. The Mississippi Delta is characterized by high rates of subsidence partially due to compaction caused by decomposition of carbon-rich Holocene era sediment deposits (Törnqvist et al. 2008), large land loss (Hartig, Kolker, and Gornitz 2002; Kolker, Allison, and Hameed 2011), and/or pores left from oil extraction (Kolker, Allison, and Hameed 2011; Mallman and Zolback 2007). The sediments of the MS Delta in Louisiana are also less compacted than sediments in most of the rest of the NGOM due to larger total porosity (Lisle and Comer 2011), and thus are more likely to be eroded by wave action (McLoughlin 2010; Fagherazzi et al. 2013; Allen 1989). Thus, increased wave height in the MS Delta watershed would lead to increased wetland loss as derived by our model, irrespective of RSLR.

In contrast, the negative effect of wave height on coastal wetland loss in the southern Florida watershed (#1) is unexpected but is likely purely correlational due to low wave height and high wetland loss in this watershed. It may be better interpreted as that coastal wetland loss was large even under low wave height (Figure 2). Mangroves in the southern Florida watershed instead of salt marshes may explain low wave height as they reduce wave energy and wave height because of mechanical friction from roots and trunks (Parvathy and Bhaskaran 2017). Mangroves loss is large in this watershed, likely due to complex anthropogenic influences beyond land use/land cover change (Goldberg et al. 2020).

It was expected that coastal slope and tidal range would lower coastal wetland loss as steeper coastal slopes act as a physical barrier to inundation and larger tidal range could deposit more ocean-borne sediment onto coastal wetlands (Kirwan and Guntenspergen 2010). The general negative medians of their coefficients for each watershed in our study support this assertion (Figure 4). However, a steeper slope does not facilitate upslope migration of salt marshes and thus may contribute to wetland loss. We did not consider this process in our model, but the wetland change may show up in the response variable. The dual effects of coastal slope may explain why it was not selected in the best model (Model 58).

NDVI did not affect coastal wetland loss shown in the models as it was either not included in the best model candidates or the posterior of its coefficient overlapped zero. This may be because NDVI may not adequately represent vegetation productivity. Although NDVI has been used in the NGOM to predict above-ground vegetation productivity and biomass (Couvillion and Beck 2013; Mishra et al. 2012; Mo, Kearney, and Momen 2017; Mo, Kearney, and Turner 2019), an evaluation of multiple remote sensing derived vegetation indices indicated a large spatial variability in the index that best represents peak green above-ground biomass (Wu et al. 2018). Some studies show wetland health is rarely correlated with above-ground biomass (Nyman et al. 1994; Turner et al. 2004; Kirwan and Guntenspergen 2012) as below-ground biomass is key to stability of coastal wetlands (Tahsin, Medeiros, and Singh 2016; Byrd et al. 2018; Turner et al. 2004; Stagg et al. 2016). The characteristics of below-ground biomass cannot be directly related to surface reflectance detected by remote sensors. Inundation levels in coastal wetlands can also affect NDVI and NDVI-based estimates of biomass (Klemas 2013; Kearney et al. 2009). Finally, our choice of the 1996 period (June – August) for NDVI, the beginning of the decadal change, may not be ideal to account for the long-term effect of vegetation status. Though vegetation cover in the prior year was the best single predictor of subsequent coastal wetland loss in coastal Louisiana (Schoolmaster et al. 2018), this does not necessarily extrapolate to a long-term and larger spatial extent investigation as done in our study. NDVI in the middle of the decade or the mean decadal NDVI may be worth exploring further.

Missing drivers

We did not select all the potential drivers in our study as illustrated by the Bayesian p-value (based on coefficient of variance) of 0.28 for the best model. We chose to focus on potentially important biogeophysical drivers that are readily available spatially. Some fine-scale covariates (i.e., site scale) that can affect coastal wetland vulnerability to RSLR are salinity, river/tide ratio (Braswell and Heffernan 2019), interior marsh inundation (Day et al. 2000), wind driven wave height (Marani et al. 2011), vegetation type, and associated flow attenuation (Rodríguez et al. 2017). In addition, storms, sediment deprivation, oil and gas extraction and infrastructure, navigation infrastructure, and saltwater intrusion altered hydrology and likely coastal wetlands (Penland et al. 2001; Couvillion et al. 2017). For example, river borne-sediment deprivation in Louisiana as indicated by a 50% decline in sediment supply since 1950s due to upland hydrological changes (Reyes et al. 2000; Blum and Roberts 2009), progressive soil waterlogging due to reduced sediment input, and interaction with salinity, produce deleterious effects on vegetation (Reyes et al. 2000; Turner 1990).

Anthropogenic factors were not explicitly accounted for in our models of wetland loss except land use/land cover (Lin and Shen 2018; Li et al. 2018), though some impacts were likely pooled into watershed boundaries, such as pollutant discharge (Lin and Shen 2018), damming, dredging and navigation activities (Turner 1997; Li et al. 2018), and population density (Entwistle, Mora, and Knight 2018). However, some human activities occur at the estuary scale (i.e., between site and watershed scales) and the addition of the estuary scale would likely improve the models further. We also did not include climate variables and soil properties directly in our models. However, climate variables directly affect vegetation growth, so the vegetation index would likely capture some of the variability of climate variables. Furthermore, RSLR may affect wave height or tidal range (Passeri et al. 2016), while their impacts to each other were not included in the current study. Finally, we used HUC4 as a balanced approach between watershed size and number of sites in each watershed, and we believe the choice of scales in our study is appropriate. However, the selection of watershed size can be more refined than HUC4 if a sufficient

number of sites are available in each watershed. More covariates like sediment availability that are relevant to the study design are important to be included as they should improve the model predictions if they are available.

Implication for future research and resource management

The Bayesian multi-level modeling approach we developed provides a flexible framework that can be easily adapted to incorporate more data while updating the posteriors with uncertainty reduced. It has the flexibility of integrating different spatial-scale data (see Figure 3, such as adding a new spatial scale, or adding covariates at an existing spatial scale like indicators for anthropogenic activities), and can be applied more broadly in other regions with different environmental settings as long as the required data on wetland loss and potential drivers are available. This modeling approach is useful in identifying the biogeophysical factors affecting wetland loss and detecting hotspots of vulnerability (tier 1). It facilitates more in-depth analysis at the hotspots of vulnerable wetlands that would require urgent attention and timely collection of more fine-scale data to understand the processes that contribute to that vulnerability (tier 2). We found we could reduce research labor costs and resources by using the two-tier research approach but the modeling framework may be limited by the lack of data for some fine-scale drivers.

In addition to increased understanding of coastal wetland dynamics, quantifying the dominant drivers for coastal wetland loss at a particular watershed, as done in our study, has important implications for coastal wetland restoration. In watersheds where dominant drivers are difficult to identify, more comprehensive restoration strategies that address multiple drivers will be needed. If RSLR is the dominant driver, sediment enhancement through beneficial use of dredged materials and/or river diversion, or increases in hydrological connectivity may be effective (Ravens et al. 2009), like in MS coast (#8), western Louisiana or Chenier Plain (#10), and MS Delta (#9) watersheds. If wave height is the main driver, breakwater structures or living shorelines to reduce wave energy may be considered to mitigate wetland loss more effectively like in MS Delta (#9) (Davis et al. 2015). Finally, the real scenarios may require multiple strategies simultaneously to conserve or restore coastal wetlands effectively due to the complexity of coastal wetland dynamics. For the MS Delta (#9), both RSLR and wave height positively (or marginally positively) affected coastal wetland loss and thus effective restoration strategies would require the combination of freshwater diversion, living shorelines and/or breakwater infrastructure (Hardy 2018).

5. Conclusions

Our study shows that the multi-level Bayesian models as an effective way to the analysis of coastal wetland change at a broad spatial scale. Although we focused on the NGOM, the modeling framework we developed can be readily applied to other regions in the US as the data we used are available at the national level. The results demonstrate the importance of considering watershed boundaries as the impacts of *in-situ* and ocean-derived biogeophysical factors on coastal wetland loss varied by watershed. The large variances of these impacts derived from these models illustrate the complexity of modeling wetland loss processes at a gulf-wide scale. Accordingly, effective restoration and conservation efforts require resource managers to develop strategies based on targeting multiple environmental drivers simultaneously. Furthermore, the research provides a cost-effective alternative to current labor and resource intensive monitoring efforts, especially through the identification of multiple dominant drivers to guide finer-scale data collection and mechanistic analysis in the future.

Acknowledgements

This work was supported by 2015 Exploratory Grant #2000005979 funded by the National Academy of Sciences through its Gulf Research Program, and by a research assistantship for the senior author from the Division of Coastal Sciences, The University of Southern Mississippi.

We would like to thank Drs. R. Leaf and D. Petrolia for their critical review on the Hardy's thesis, one chapter of which leads to this manuscript. We would also like to thank our colleagues at the USGS (Dr. H. Wang and M. Fisher) and those who made data available for use at the USGS, NASA, and big data

computation accesssible at Amazon Cloud Services (AWS), and Google Earth Engine.

Data Availability Statement

Data and code for analysis can be found at the project's GitHub page at http://github.com/ecospatial/NAS_2016. We have also created a READ.ME file for users and the complete data archiving is underway.

Disclosure conflicts of interest statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Gulf Research Program of the National Academies [2000005979].

ORCID

Tyler Hardy http://orcid.org/0000-0002-9087-2915
Wei Wu http://orcid.org/0000-0002-7945-9847
Mark S. Peterson http://orcid.org/0000-0002-2853-8182

References

Alber, M. 2002. "A Conceptual Model of Estuarine Freshwater Inflow Management." *Estuaries* 25 (6): 1246–1261. doi:10.1007/BF02692222.

Alizad, K., S. C. Hagen, S. C. Medeiros, M. V. Bilskie, J. T. Morris, L. Balthis, and C. A. Buckel. 2018. "Dynamic Responses and Implications to Coastal Wetlands and the Surrounding Regions under Sea Level Rise." *PLoS ONE* 13 (10): e0205176. doi:10.1371/journal.pone.0205176.

Alizad, K., S. C. Hagen, J. T. Morris, P. Bacopoulos, M. V. Bilskie, J. F. Weishampel, and S. C. Medeiros. 2016. "A Coupled, Two-Dimensional Hydrodynamic-Marsh Model with Biological Feedback." *Ecological Modelling* 327: 29–43. Elsevier B.V. doi:10.1016/j.ecolmodel.2016.01.013.

Allen, J. R. L. 1989. "Evolution of Salt-Marsh Cliffs in Muddy and Sandy Systems: A Qualitative Comparison of British West-Coast Estuaries." *Earth Surface Processes and Landforms* 14 (1): 85–92. doi:10.1002/esp.3290140108.

Barbier, E. B. 2013. "Valuing Ecosystem Services for Coastal Wetland Protection and Restoration: Progress and Challenges." *Resources* 2 (3): 213–230. doi:10.3390/resources2030213.

Barbier, E. B. 2016. "The Protective Value of Estuarine and Coastal Ecosystem Services in a Wealth Accounting Framework." *Environmental and Resource Economics* 64 (1): 37–58. doi:10.1007/s10640-015-9931-z.

Blum, M. D., and H. H. Roberts. 2009. "Drowning of the Mississippi Delta Due to Insufficient Sediment Supply and Global Sea-Levelrise." *Nature Geoscience* 2 (7): 488–491. doi:10.1038/ngeo553.

Blum, M. D., and H. H. Roberts. 2012. "The Mississippi Delta Region: Past, Present, and Future." *Annual Review of Earth and Planetary Sciences* 40 (1): 655–683. doi:10.1146/annurev-earth-042711-105248.

Bolker, B. M., M. E. Brooks, C. J. Clark, S. W. Geange, J. R. Poulsen, M. Henry, H. Stevens, J. Simone, and S. White. 2009. "Generalized Linear Mixed Models: A Practical Guide for Ecology and Evolution." *Trends in Ecology & Evolution* 24 (3): 127–135. doi:10.1016/j.tree.2008.10.008.

Borchert, S. M., M. J. Osland, N. M. Enwright, and K. T. Griffith. 2018. "Coastal Wetland Adaptation to Sea Level Rise: Quantifying Potential for Landward Migration and Coastal Squeeze." *Journal of Applied Ecology*, 1: 1–12. doi:10.1111/1365-2664.13169.

Bourne, J. 2000. "Louisiana 'S Vanishing Wetlands : Going, Going." *Science* 289 (5486): 1860–1863. doi:10.1126/science.289.5486.1860.

Braswell, A. E., and J. B. Heffernan. 2019. "Coastal Wetland Distributions: Delineating Domains of Macroscale Drivers and Local Feedbacks." *Ecosystems* 22 (6): 1256–1270. doi:10.1007/s10021-018-0332-3. Springer US.

Burnham, K. P., D. R. Anderson, and K. P. Huyvaert. 2011. "AIC Model Selection and Multimodel Inference in Behavioral Ecology: Some Background, Observations, and Comparisons." *Behavioral Ecology and Sociobiology* 65 (1): 23–35. doi:10.1007/s00265-010-1029-6.

Burnham, K. P., and R. P. Anderson. 2004. "Multimodel Inference: Understanding AIC and BIC in Model Selection." *Sociological Methods & Research* 33 (2): 261–304. doi:10.1177/0049124104268644.

Byrd, K. B., L. Ballanti, N. Thomas, J. R. Dung Nguyen, M. S. Holmquist, and L. Windham-Myers. 2018. "A Remote Sensing-Based Model of Tidal Marsh Aboveground Carbon Stocks for the Conterminous United States." *ISPRS Journal of Photogrammetry and Remote Sensing* 139: 255–271. doi:10.1016/j.isprsjprs.2018.03.019.

Caffey, R. H., H. Wang, and D. R. Petrolia. 2014. "Trajectory Economics: Assessing the Flow of Ecosystem Services from Coastal Restoration." *Ecological Economics* 100: 74–84. doi:10.1016/j.ecolecon.2014.01.011.

Calder, C., M. Lavine, P. Müller, J. S. S. Clark, P. Muller, and J. S. S. Clark. 2003. "Incorporating Multiple Sources of Stochasticity into Dynamic Population Models." *Ecology* 84 (6): 1395–1402. doi:10.1890/0012-9658(2003)084[1395:IMSOSI]2.0.CO;2.

Clark, J. S. 2001. "Ecological Forecasts: An Emerging Imperative." *Science* 293 (5530): 657–660. doi:10.1126/science.293.5530.657.

Clark, J. S. 2005. "Why Environmental Scientists Are Becoming Bayesians." *Ecology Letters* 8 (1): 2–14. doi:10.1111/j.1461-0248.2004.00702.x.

Clark, J. S., and A. E. Gelfand. 2006. "A Future for Models and Data in Environmental Science." *Trends in Ecology & Evolution* 21 (7): 375–380. doi:10.1016/j.tree.2006.03.016.

Costanza, R., M. Octavio Pérez-Maqueo, L. Martinez, P. Sutton, S. J. Anderson, and K. Mulder. 2008. "The Value of Coastal Wetlands for Hurricane Protection." *AMBIO: A Journal of the Human Environment* 37 (4): 241–248. doi:10.1579/0044-7447(2008)37[241:TVOCWF]2.0.CO;2.

Couvillion, B. R., and H. Beck. 2013. "Marsh Collapse Thresholds for Coastal Louisiana Estimated Using Elevation and Vegetation Index Data." *Journal of Coastal Research* 63: 58–67. doi:10.2112/SI63-006.1.

Couvillion, B. R., H. Beck, D. Schoolmaster, and M. Fischer. 2017. "Land Area Change in Coastal Louisiana (1932 to 2016)." *U.S. Geological Survey Scientific Investigations Map* 3381: 16. Reston, VA. doi:10.3133/sim3381.

Cowles, M. K., and B. P. Carlin. 1996. "Markov Chain Monte Carlo Convergence Diagnostics: A Comparative Review." *Journal of the American Statistical Association* 91 (434): 883–904. doi:10.1080/01621459.1996.10476956.

Dahl, T. E., and S. M. Stedman. (Eds). 2006. *Status and Trends of Wetlands in the Coastal Watersheds of the Conterminous United States 2004 to 2009*. D.C, U.S: Fish and Wildlife Service and National Oceanic and Atmospheric Administration, National Marine Fisheries Service, Department of the Interior.

Daniels, M. J., A. S. Chatterjee, and C. Wang. 2012. "Bayesian Model Selection for Incomplete Data Using the Posterior Predictive Distribution." *Biometrics* 68 (4): 1055–1063. doi:10.1111/j.1541-0420.2012.01766.x.

Davidson, E. A., and I. A. Janssens. 2006. "Temperature Sensitivity of Soil Carbon Decomposition and Feedbacks to Climate Change." *Nature* 440 (7081): 165–173. doi:10.1038/nature04514. Nature Publishing Group.

Davis, J. L., C. A. Currin, O. Colleen, C. Raffenburg, and A. Davis. 2015. "Living Shorelines: Coastal Resilience with a Blue Carbon Benefit." *PLoS ONE* 10 (11): 1–18. doi:10.1371/journal.pone.0142595.

Day, J. W., L. D. Britsch, S. R. Hawes, G. P. Shaffer, D. J. Reed, and D. Cahoon. 2000. "Pattern and Process of Land Loss in the Mississippi Delta: A Spatial and Temporal Analysis of Wetland Habitat Change." *Estuaries* 23 (4): 425–438. doi:10.2307/1353136.

Dick, F., and H. Tevaearai. 2015. "Significance and Limitations of the P Value." *European Journal of Vascular and Endovascular Surgery : The Official Journal of the European Society for Vascular Surgery* 50 (6): 815. doi:10.1016/j.ejvs.2015.07.026.

Elsey-Quirk, T., S. A. Graham, I. A. Mendelssohn, G. Snedden, J. W. Day, R. Twilley, G. Shaffer, L. A. Sharp, J. Pahl, and R. R. Lane. 2019. "Mississippi River Sediment Diversions and Coastal Wetland Sustainability: Synthesis of Responses to Freshwater, Sediment, and Nutrient Inputs." *Estuarine, Coastal and Shelf Science* 221: 170–183. doi:10.1016/j.ecss.2019.03.002.

Engle, V. D. 2011. "Estimating the Provision of Ecosystem Services by Gulf of Mexico Coastal Wetlands." *Wetlands* 31 (1): 179–193. doi:10.1007/s13157-010-0132-9.

Entwistle, C., M. A. Mora, and R. Knight. 2018. "Estimating Coastal Wetland Gain and Losses in Galveston County and Cameron County, Texas, USA." *Integrated Environmental Assessment and Management* 14 (1): 120–129. doi:10.1002/ieam.1973.

Enwright, N. M., K. T. Griffith, and M. J. Osland. 2016. "Barriers to and Opportunities for Landward Migration of Coastal Wetlands with Sea-Level Rise." *Frontiers in Ecology and the Environment* 14 (6): 307–316. doi:10.1002/fee.1282.

Fagherazzi, S., M. L. Kirwan, S. M. Mudd, G. R. Guntenspergen, S. Temmerman, D. Andrea, and V. D. K. Johan. 2012. "Numerical Models of Salt Marsh Evolution: Ecological, Geormorphic, and Climatic Factors." *Reviews of Geophysics* 50 (2011): 1–28. doi:10.1029/2011RG000359.

Fagherazzi, S., G. Mariotti, P. L. Wiberg, and K. J. McGlathery. 2013. "Marsh Collapse Does Not Require Sea Level Rise." *Oceanography* 26 (3): 70–77. doi:10.5670/oceanog.2009.80.COPYRIGHT.

Farber, S. C., R. Costanza, and M. A. Wilson. 2002. "Economic and Ecological Concepts for Valuing Ecosystem Services." *Ecological Economics* 41 (3): 375–392. doi:10.1016/S0921-8009(02)00088-5.

Ganju, N. K., Z. Defne, M. L. Kirwan, S. Fagherazzi, D. Andrea, and L. Carniello. 2017. "Spatially Integrative Metrics Reveal Hidden Vulnerability of Microtidal Salt Marshes." *Nature Communications* 8. doi:10.1038/ncomms14156.

Gelfand, A. E., and S. K. Ghosh. 1998. "Model Choice : A Minimum Posterior Predictive Loss Approach." *Biometrika* 85 (1): 1–11. doi:10.1093/biomet/85.1.1.

Gelfand, A. E., and A. F. M. Smith. 1990. "Sampling-Based Approaches to Calculating Marginal Densities." *Journal of the American Statistical Association* 85 (410): 398–409. doi:10.1080/01621459.1990.10476213.

Gelman, A., and J. Hill. 2007. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge: Cambridge University Press.

Georgiou, I. Y., D. M. FitzGerald, and G. W. Stone. 2005. "The Impact of Physical Processes along the Louisiana Coast." *Journal of Coastal Research* 44: 72–89.

Geselbracht, L. L., K. Freeman, A. P. Birch, J. Brenner, and D. R. Gordon. 2015. "Modeled Sea Level Rise Impacts on Coastal Ecosystems at Six Major Estuaries on Florida's Gulf Coast: Implications for Adaptation Planning." *PLOS ONE* 10 (7): e0132079. doi:10.1371/journal.pone.0132079.

Gilks, W. R., S. Richardson, and D. J. Spiegelhalter. 1996. "Introducing Markov Chain Monte Carlo." In *Markov Chain Monte Carlo in Practice*, edited by W. R. Gilks, S. Richardson, and D. Spiegelhalter, 1–19. London, UK: Chapman and Hall.

Glick, P., J. Clough, A. Polaczyk, B. Couvillion, and B. Nunley. 2013. "Potential Effects of Sea-Level Rise on Coastal Wetlands in Southeastern Louisiana." *Journal of Coastal Research* 63: 211–233. doi:10.2112/SI63-0017.1.

Goldberg, L., D. Lagomasino, N. Thomas, and T. Fatoyinbo. 2020. "Global Declines in Human-Driven Mangrove Loss."

Global Change Biology 26 (10): 5844–5855. doi:10.1111/gcb.15275.

Groffman, P. M., J. S. Baron, T. Blett, A. J. Gold, I. Goodman, and L. H. Gunderson, others. 2006. "Ecological Thresholds: The Key to Successful Environmental Management or an Important Concept with No Practical Application?." *Ecosystems* 9(1):1–13. doi:10.1007/s10021-003-0142-z.

Groot, D., B. R. Bernard, L. van der Ploeg, S. Costanza, B. R. Bernard, and F. L. Braat. 2012. "Global Estimates of the Value of Ecosystems and Their Services in Monetary Units." *Ecosystem Services* 1 (1): 50–61. doi:10.1016/j.ecoser.2012.07.005. Elsevier.

Gutierrez, B. T., N. G. Plant, and E. R. Thieler. 2011a. "A Bayesian Network to Predict Coastal Vulnerability to Sea Level Rise." *Journal of Geophysical Research: Earth Surface* 116: F02009. doi:10.1029/2010JF001891.

Gutierrez, B. T., N. G. Plant, and E. R. Thieler. 2011b. "A Bayesian Network to Predict Vulnerability to Sea-Level Rise: Data Report," 1–15.

Hagen, S. C., J. T. Morris, P. Bacopoulos, and J. F. Weishampel. 2012. "Sea-Level Rise Impact on a Salt Marsh System of the Lower St. Johns River." *Journal of Waterway, Port, Coastal, and Ocean Engineering* 139 (2): 118–125. doi:10.1061/(ASCE)WW.1943-5460.0000177.

Hardisky, M. A., M. F. Gross Gross, and V. Klemas. 1986. "Remote Sensing of Coastal Wetlands." *Bioscience* 36 (7): 453–460. doi:10.2307/1310341.

Hardy, T. 2018. *Coastal Wetland Dynamics under Sea-Level Rise and Wetland Restoration in the Northern Gulf of Mexico Using Bayesian Multilevel Models and a Web Tool*. Hattiesburg, Mississippi: University of Southern Mississippi.

Hartig, E. K., A. Kolker, and V. Gornitz. 2002. "Investigations into Recent Salt Marsh Losses in Jamaica Bay, New York." http://www.ldeo.columbia.edu/res/fac/physocean/Jambay/SaltMarshes.pdf .

Hobbs, N. T., and M. B. Hooten. 2016. *Bayesian Models: A Statistical Primer for Ecologists*. Princeton, New Jersey: Princeton University Press.

Hooten, M. B., and N. T. Hobbs. 2015. "A Guide to Bayesian Model Selection for Ecologists." *Ecological Monographs* 85 (1): 3–28. doi:10.1890/14-0661.1.

Hovenga, P. A., D. Wang, S. C. Medeiros, S. C. Hagen, and K. Alizad. 2016. "The Response of Runoff and Sediment Loading in the Apalachicola River, Florida to Climate and Land Use Land Cover Change." *Earth's Futures Future* 45 (5): 124–142. doi:10.1002/2015EF000348.

Huang, W., S. C. Hagen, D. Wang, P. A. Hovenga, F. Teng, and J. F. Weishampel. 2016. "Suspended Sediment Projections in Apalachicola Bay in Response to Altered River Flow and Sediment Loads under Climate Change and Sea Level Rise." *Earth's Futures Future* 4 (10): 428–439. doi:10.1002/2016EF000384.

Ibáñez, I., J. A. S. J. Silander, A. M. Wilson, N. LaFleur, N. N. Tanaka, and I. Tsuyama, others. 2009. Multivariate Forecast of Potential Distributions of Invasive Species. *Ecological Applications* 19 (2): 359–375. doi:10.1890/07-2095.1.

Interis, M. G., and D. R. Petrolia. 2016. "Location, Location, Habitat: How the Value of Ecosystem Services Varies across Location and by Habitat." *Land Economics* 92 (2): 292–307. http://le.uwpress.org/content/by/year. doi:10.3368/le.92.2.292.

Jankowski, K. L., T. E. Törnqvist, A. M. Fernandes, T. E. T?rnqvist, A. M. Fernandes, and T. E. T?rnqvist. 2017. "Vulnerability of Louisiana's Coastal Wetlands to Present-Day Rates of Relative Sea-Level Rise." *Nature Communications* 8 (1): 14792. doi:10.1038/ncomms14792.

Karegar, M. A., T. H. Dixon, and R. Malservisi. 2015. "A Three-Dimensional Surface Velocity Field for the Mississippi Delta: Implications for Coastal Restoration and Flood Potential." *Geology* 43 (6): 519–522. doi:10.1130/G36598.1.

Karimpour, A., Q. Chen, and R. R. Twilley. 2016. "A Field Study of How Wind Waves and Currents May Contribute to the Deterioration of Saltmarsh Fringe." *Estuaries and Coasts* 39: 935–950. doi:10.1007/s12237-015-0047-z.

Kearney, M. S., D. Stutzer, K. Turpie, and J. Court Stevenson. 2009. "The Effects of Tidal Inundation on the Reflectance Characteristics of Coastal Marsh Vegetation." *Journal of Coastal Research* 25 (6): 1177–1186. doi:10.2112/08-1080.1.

Kennedy, C. J., and E. B. Barbier. 2016. "The Economic Value of Freshwater Inputs to an Estuarine Fishery." *Water Resources and Economics* 13: 46–59. Elsevier. doi:10.1016/j.wre.2015.11.003.

Kéry, M. 2010. *Introduction to Winbugs for Ecologists: Bayesian Approach to Regression, Anova, Mixed Models and Related Analyses*. Burlington, MA: Academic Press.

Kidwell, D. M., J. C. Dietrich, S. C. Hagen, and S. C. Medeiros. 2017. "An Earth's Future Special Collection: Impacts of the Coastal Dynamics of Sea Level Rise on Low-Gradient Coastal Landscapes." *Earth's Future* 5 (1): 2–9. doi:10.1002/2016EF000493.

Kirwan, M. L., and A. Brad Murray. 2007. "A Coupled Geomorphic and Ecological Model of Tidal Marsh Evolution." *Proceedings of the National Academy of Sciences of the United States of America PNAS*, 104 (15): 6118–6122. doi:10.1073/pnas.0700958104.

Kirwan, M. L., and G. R. Guntenspergen. 2009. "Accelerated Sea-level Rise – A Response to Craft Et Al." *Frontiers in Ecology and the Environment* (3). doi:10.1890/09.WB.005.

Kirwan, M. L., and G. R. Guntenspergen. 2010. "Influence of Tidal Range on the Stability of Coastal Marshland." *Journal of Geophysical Research: Earth Surface* 115: F02009. doi:10.1029/2009JF001400.

Kirwan, M. L., and G. R. Guntenspergen. 2012. "Feedbacks between Inundation, Root Production, and Shoot Growth in a Rapidly Submerging Brackish Marsh." *Journal of Ecology* 100 (3): 764–770. doi:10.1111/j.1365-2745.2012.01957.x.

Kirwan, M. L., and J. Patrick Megonigal. 2013. "Tidal Wetland Stability in the Face of Human Impacts and Sea-Level Rise." *Nature* 504 (7478): 53–60. doi:10.1038/nature12856.

Klemas, V. 2013. "Remote Sensing of Coastal Wetland Biomass: An Overview." *Journal of Coastal Research* 290: 1016–1028. doi:10.2112/JCOASTRES-D-12-00237.1.

Kolker, A. S., M. A. Allison, and S. Hameed. 2011. "An Evaluation of Subsidence Rates and Sea-Level Variability in the

Northern Gulf of Mexico." *Geophysical Research Letters* 38 (21): L21404. doi:10.1029/2011GL049458.

LaFever, D. H., R. R. Lopez, R. A. Feagin, and N. J. Silvy. 2007. "Predicting the Impacts of Future Sea-Level Rise on an Endangered Lagomorph." *Environmental Management* 40 (3): 430–437. doi:10.1007/s00267-006-0204-z.

Lambert, P. C., A. J. Sutton, P. R. Burton, K. R. Abrams, and D. R. Jones. 2005. "How Vague Is Vague? A Simulation Study of the Impact of the Use of Vague Prior Distributions in MCMC Using WinBUGS." *Statistics in Medicine* 24 (15): 2401–2428. doi:10.1002/sim.2112.

Li, X., R. Bellerby, C. Craft, and S. E. Widney. 2018. "Coastal Wetland Loss, Consequences, and Challenges for Restoration." *Anthropocene Coasts* 15 (January): 1–15. doi:10.1139/anc-2017-0001.

Lin, Q., and Y. Shen. 2018. "Losses of Natural Coastal Wetlands by Land Conversion and Ecological Degradation in the Urbanizing Chinese Coast." *Scientific Reports* 8 (1): 1–10. doi:10.1038/s41598-018-33406-x.

Linhoss, A. C., G. Kiker, M. Shirley, and K. Frank. 2015. "Sea-Level Rise, Inundation, and Marsh Migration: Simulating Impacts on Developed Lands and Environmental Systems." *Journal of Coastal Research* 299 (1): 36–46. doi:10.2112/JCOASTRES-D-13-00215.1.

Lisle, J. T., and N. N. Comer. 2011. *Characterization of Sediments from the Gulf of Mexico and Atlantic Shorelines, Texas to Florida. Open-File Report 2011–1199*. Reston, Virginia. https://pubs.usgs.gov/of/2011/1199/pdf/OFR-2011-1199.pdf .

Mallman, E. P., and M. D. Zolback. 2007. "Subsidence in the Louisiana Coastal Zone Due to Hydrocarbon Production." *Journal of Coastal Research Special Is* 50: 443–448.

Marani, M., D. Andrea, S. Lanzoni, L. Carniello, and A. Rinaldo. 2007. "Biologically-Controlled Multiple Equilibria of Tidal Landforms and the Fate of the Venice Lagoon." *Geophysical Research Letters* 34 (11): 1–5. doi:10.1029/2007GL030178.

Marani, M., A. D'Alpaos, S. Lanzoni, and M. Santalucia. 2011. "Understanding and Predicting Wave Erosion of Marsh Edges." *Geophysical Research Letters* 38 (21): 1–5. doi:10.1029/2011GL048995.

Mariotti, G., and S. Fagherazzi. 2013. "Critical Width of Tidal Flats Triggers Marsh Collapse in the Absence of Sea-Level Rise." *Proceedings of the National Academy of Sciences PNAS*, 110 (14), 5353–5356. doi:10.1073/pnas.1219600110.

Markley, J. L., C. McMillan, and G. A. Jr Thompson. 1982. "Latitudinal Differentiation in Response to Chilling Temperatures among Populations of Three Mangroves, Avicennia Germinans, Laguncularia Racemosa, and Rhizophora Mangle, from the Western Tropical Atlantic and Pacific Panama." *Canadian Journal of Botany* 60 (12): 2704–2715. doi:10.1139/b82-330.

Martin, J. F., and M. L. White. 2000. "Evaluation of Coastal Management Plans with a Spatial Model : Mississippi Delta, Louisiana, USA." *Environmental Management* 26 (2): 117–129. doi:10.1007/s002670010075.

McLoughlin, S. M. 2010. *Erosional Processes along Salt Marsh Edges on the Eastern Shore of Virginia Thesis*. Charlottesville, VA: University of Virginia.

Meade, R. H., and J. A. Moody. 2010. "Causes for the Decline of Suspended-Sediment Discharge in the Mississippi River System, 1940-2007." *Hydrological Processes* 24: 35–49.

Mendelssohn, I. A., G. L. Andersen, D. M. Baltz, R. H. Caffey, K. R. Carman, and J. W. Fleeger, others. 2012. Oil Impacts on Coastal Wetlands: Implications for the Mississippi River Delta Ecosystem after the Deepwater Horizon Oil Spill. *BioScience* 62 (6): 562–574. doi:10.1525/bio.2012.62.6.7.

Mendelssohn, I. A., M. R. Byrnes, R. T. Kneib, and B. A. Vittor. 2017. "Coastal Habitats of the Gulf of Mexico." In *Habitats and Biota of the Gulf of Mexico: Before the Deepwater Horizon Oil Spill*, edited by C. H. Ward, 359–640. New York, NY: Springer.

Meng, X. L. 1994. "Posterior Predictive P-Values." *Annals of Statistics* 22 (3): 1142–1160. doi:10.1214/aos/1176325622.

Mishra, D. R., H. J. Cho, S. Ghosh, A. Fox, C. Downs, P. B. T. Merani, P. Kirui, N. Jackson, and S. Mishra. 2012. "Post-Spill State of the Marsh: Remote Estimation of the Ecological Impact of the Gulf of Mexico Oil Spill on Louisiana Salt Marshes." *Remote Sensing of Environment* 118: 176–185. Elsevier Inc. doi:10.1016/j.rse.2011.11.007.

Mo, Y., M. Kearney, and B. Momen. 2017. "Drought-Associated Phenological Changes of Coastal Marshes in Louisiana." *Ecosphere* 8 (5): e01811. doi:10.1002/ecs2.1811.

Mo, Y., M. S. Kearney, and R. E. Turner. 2019. "Feedback of Coastal Marshes to Climate Change: Long-Term Phenological Shifts." *Ecology and Evolution* 9 (12): 6785–6797. doi:10.1002/ece3.5215.

Morris, J. T., P. V. Sundareshwar, C. T. Nietch, B. Kjerfve, and D. R. Cahoon. 2002. "Responses of Coastal Wetlands to Rising Sea Level." *Ecology* 83 (10): 2869–2877. doi:10.1890/0012-9658(2002)083[2869:ROCWTR]2.0.CO;2.

Mudd, S. M., S. Fagherazzi, J. T. Morris, and D. J. Furbish. 2013. "Flow, Sedimentation, and Biomass Production on a Vegetated Salt Marsh in South Carolina: Toward a Predictive Model of Marsh Morphologic and Ecologic Evolution." In *Coastal and Estuarine Studies 59: The Ecogeomorphology of Tidal Marshes*, edited by S. Fagherazzi, 165-288. American Geophysical Union, USA. doi:10.1029/CE059p0165.

Mudd, S. M., S. M. Howell, and J. T. Morris. 2009. "Impact of Dynamic Feedbacks between Sedimentation, Sea-Level Rise, and Biomass Production on near-Surface Marsh Stratigraphy and Carbon Accumulation." *Estuarine, Coastal and Shelf Science* 82 (3): 377–389. doi:10.1016/j.ecss.2009.01.028.

Nyman, J. A., M. Carloss, R. D. DeLaune, and W. H. Patrick Jr. 1994. "Erosion Rather than Plant Dieback as the Mechanism of Marsh Loss in an Estuarine Marsh." *Earth Surface Processes and Landforms* 19 (1): 69–84. doi:10.1002/esp.3290190106.

Nyman, J. A., R. J. Walters, R. D. Delaune, and W. H. Patrick. 2006. "Marsh Vertical Accretion via Vegetative Growth." *Estuarine, Coastal and Shelf Science* 69 (3–4): 370–380. doi:10.1016/j.ecss.2006.05.041.

Osland, M. J., N. Enwright, R. H. Day, and T. W. Doyle. 2013. "Winter Climate Change and Coastal Wetland Foundation Species: Salt Marshes Vs. Mangrove Forests in the Southeastern United States." *Global Change Biology* 19 (5): 1482–1494. doi:10.1111/gcb.12126.

Parvathy, K. G., and P. K. Bhaskaran. 2017. "Wave Attenuation in Presence of Mangroves: A Sensitivity Study for Varying Bottom Slopes." *The International Journal of Ocean and Climate Systems* 8: 126-134. doi:10.1177/1759313117702919.

Passeri, D. L., S. C. Hagen, N. G. Plant, M. V. Bilskie, S. C. Medeiros, and K. Alizad. 2016. "Tidal Hydrodynamics under Future Sea Level Rise and Coastal Morphology in the Northern Gulf of Mexico." *Earth's Futures Future* 4 (12): 159–176. doi:10.1002/2015EF000332.

Passeri, D. L., S. C. Hagen, S. C. Medeiros, M. V. Bilskie, K. Alizad, and D. Wang. 2015. "Earth's Future Special Section : The Dynamic Effects of Sea Level Rise on Low-Gradient Coastal Landscapes : A Review." *Earth's Future* 3 (6): 159–181. doi:10.1002/2015EF000298.

Payero, J. O., C. M. U. Neale, and J. L. Wright. 2004. "Comparison of Eleven Vegetation Indices for Estimating Plant Height of Alfalfa and Grass." *Applied Engineering in Agriculture* 20 (3): 385–393. doi:10.1017/CBO9781107415324.004.

Penland, S., L. Wayne, L. D. Britsch, S. J. William, A. D. Beall, and V. C. Butterworth. 2001. *Geomorphic Classification of Coastal Land Loss between 1932 and 1990 in the Mississippi River Delta Plain, Southeastern Louisiana*. USGS Open-File Report 2000-417. Available at https://pubs.er.usgs.gov/publication/ofr00417

Peters, D. P. C., P. M. Groffman, K. J. Nadelhoffer, N. B. Grimm, S. L. Collins, W. K. Michener, and M. A. Huston. 2008. "Living in an Increasingly Connected World: A Framework for Continental-Scale Environmental Science." *Frontiers in Ecology and the Environment* 6 (5): 229–237. doi:10.1890/070098.

Petrolia, D. R., M. G. Interis, J. Hwang, D. R. Petrolia, M. G. Interis, and J. Hwang. 2014. "America S Wetland? A National Survey of Willingness to Pay for Restoration of Louisiana's Coastal Wetlands." *Marine Resource Economics* 29 (1): 17–37. doi:10.1086/676289.

Plummer, M. 2016. "Rjags: Bayesian Graphical Models Using MCMC." https://cran.r-project.org/package=rjags .

Qian, S. S., T. F. Cuffney, I. Alameddine, G. McMahon, and K. H. Reckhow. 2010. "On the Application of Multilevel Modeling in Environmental and Ecological Studies." *Ecology* 91 (2): 355–361. doi:10.1890/09-1043.1.

R Core Team. 2015. "R: A Language and Environment for Statistical Computing." Vienna, Austria, Austria: R Foundation for Statistical Computing. https://www.r-project.org/ .

Ravens, T. M., R. C. Thomas, K. A. Roberts, P. H. Santschif, and P. H. Santschi. 2009. "Causes of Salt Marsh Erosion in Galveston Bay, Texas." *Journal of Coastal Research* 25 (2): 265–272. doi:10.2112/07-0942.1.

Reed, D. J. J. 1995. "The Response of Coastal Marshes to Sea-Level Rise: Survival of Submergence?." *Earth Surface Processes and Landforms* 20 (1): 39–48. doi:10.1002/esp.3290200105.

Reyes, E., M. L. White, J. F. Martin, G. Paul Kemp, J. W. Day, and V. Aravamuthan. 2000. "Landscape Modeling of Coastal Habitat Change in the Mississippi Delta." *Ecology* 81 (8): 2331–2349. doi:10.1890/0012-9658(2000)081[2331:LMOCHC]2.0.CO;2.

Park, Richard, A. R. A., S. M. S. Manjit, R. C. Trehan, R. C. Howe, P. W. P. W. Mausel, R. C. Robert, and C. Howe. 1989. "The Effects of Sea Level Rise on U.S. Coastal Wetlands April 1989." In*The Potential Effects of Global Climate Change on the United States, Appendix B - Sea Level Rise*, edited by J. B. Smith and D. A. Tirpak, 56. Washington, DC: U.S. Environmental Protection Agency.

Thieler, E. R., and E. S. Hammar-Klose. 1999. *National Assessment of Coastal Vulnerability to Sea-Level Rise: Preliminary Results for the U.A. Atlantic Coast*. https://pubs.usgs.gov/of/1999/of99-593/ .

Rodríguez, J. F., P. M. Saco, S. Sandi, N. Saintilan, and G. Riccardi. 2017. "Potential Increase in Coastal Wetland Vulnerability to Sea-Level Rise Suggested by considering Hydrodynamic Attenuation Effects." *Nature Communications* 8 (1): 16094. doi:10.1038/ncomms16094.

Rogers, K., N. Saintilan, and C. Copeland. 2012. "Modelling Wetland Surface Elevation Dynamics and Its Application to Forecasting the Effects of Sea-Level Rise on Estuarine Wetlands." *Ecological Modelling* 244: 27–36. doi:10.1016/j.ecolmodel.2013.04.016.

Ross, M. S., J. J. O'Brien, R. Glenn Ford, K. Zhang, and A. Morkill. 2009. "Disturbance and the Rising Tide: The Challenge of Biodiversity Management on Low-Island Ecosystems." *Frontiers in Ecology and the Environment* 7 (9): 471–478. doi:10.1890/070221.

Schile, L. M., J. C. Callaway, J. T. Morris, D. Stralberg, V. Thomas Parker, and M. Kelly. 2014. "Modeling Tidal Marsh Distribution with Sea-Level Rise: Evaluating the Role of Vegetation, Sediment, and Upland Habitat in Marsh Resiliency." *PLoS ONE* 9(2): e88760. doi:10.1371/journal.pone.0088760.

Schoolmaster, D. R., C. L. Stagg, L. A. Sharp, T. E. McGinnis, B. Wood, and S. C. Piazza. 2018. "Vegetation Cover, Tidal Amplitude and Land Area Predict Short-Term Marsh Vulnerability in Coastal Louisiana." *Ecosystems* 21 (7): 1335–1347. doi:10.1007/s10021-018-0223-7.

Simas, T., J. P. Nunes, and J. G. Ferreira. 2001. "Effects of Global Climate Change on Coastal Salt Marshes." *Ecological Modelling* 139 (1): 1–15. doi:10.1016/S0304-3800(01)00226-5.

Stagg, C. L., K. W. Krauss, D. R. Cahoon, N. Cormier, W. H. Conner, and C. M. Swarzenski. 2016. "Processes Contributing to Resilience of Coastal Wetlands to Sea-Level Rise." *Ecosystems* 19 (8): 1445–1459. doi:10.1007/s10021-016-0015-x.

Stralberg, D., M. Brennan, J. C. Callaway, J. K. Wood, L. M. Schile, D. Jongsomjit, M. Kelly, V. Thomas Parker, and S. Crooks. 2011. "Evaluating Tidal Marsh Sustainability in the Face of Sea-Level Rise: A Hybrid Modeling Approach Applied to San Francisco Bay." *PLoS ONE* 6(11): e27388. doi:10.1371/journal.pone.0027388.

Syvitski, J. P. M., C. J. Vörösmarty, A. J. Kettner, and P. Green. 2005. "Impact of Humans on the Flux of Terrestrial Sediment

to the Global Coastal Ocean." *Science* 308 (5720): 376–380. doi:10.1126/science.1109454.

Tahsin, S., S. C. Medeiros, and A. Singh. 2016. "Resilience of Coastal Wetlands to Extreme Hydrologic Events in Apalachicola Bay." *Geophysical Research Letters* 43 (14): 7529–7537. doi:10.1002/2016GL069594.

Technow, F., and L. Radu Totir. 2015. "Using Bayesian Multilevel Whole Genome Regression Models for Partial Pooling of Training Sets in Genomic Prediction." *Genes|Genomes|Genetics* 5 (8): 1603–1612. doi:10.1534/g3.115.019299.

Templet, P. H., and K. J. Meyer-Arendt. 1988. "Louisiana Wetland Loss: A Regional Water Management Approach to the Problem." *Environmental Management* 12 (2): 181–192. doi:10.1007/BF01873387.

Törnqvist, T. E., D. J. Wallace, J. E. A. Storms, J. Wallinga, R. L. Van Dam, and M. Blaauw, others. 2008. Mississippi Delta Subsidence Primarily Caused by Compaction of Holocene Strata. *Nature Geoscience* 1 (3): 173–176. doi:10.1038/ngeo129.

Turner, R. E. 1990. "Landscape Development and Coastal Wetland Losses in the Northern Gulf of Mexico." *American Zoologist* 30 (1): 89–105. doi:10.1093/icb/30.1.89.

Turner, R. E., and D. R. Cahoon. 1987. *Causes of Wetland Loss in the Coastal Central Gulf of Mexico. Volume II: Technical Narrative. Final Report Submitted to Minerals Management Service, New Orleans, LA. Contract NO. 14- 12-0001-30252. OCS Study/MMS 87-0120. 400pp*. Vol. II. http://repositories.tdl.org/tamug-ir/handle/1969.3/21213 .

Turner, R. E., E. M. Swenson, C. S. Milan, J. M. Lee, and T. A. Oswald. 2004. "Below-Ground Biomass in Healthy and Impaired Salt Marshes." *Ecological Research* 19 (1): 29–35. doi:10.1111/j.1440-1703.2003.00610.x.

Turner, R. E. 1997. "Wetland Loss in the Northern Gulf of Mexico: Multiple Working Hypotheses." *Estuaries* 20 (1): 1–13. doi:10.2307/1352716.

Tweel, A. W., and R. E. Turner. 2012. "Landscape-Scale Analysis of Wetland Sediment Deposition from Four Tropical Cyclone Events." *PLoS ONE* 7 (11). doi:10.1371/journal.pone.0050528.

Vörösmarty, C. J., M. Meybeck, B. Fekete, K. Sharma, P. Green, and J. P. M. Syvitski. 2003. "Anthropogenic Sediment Retention: Major Global Impact from Registered River Impoundments." *Global and Planetary Change* 39 (1–2): 169–190. doi:10.1016/S0921-8181(03)00023-7.

Walker, H., J. Coleman, H. Roberts, and R. Tye. 1987. "Wetland Loss in Louisiana." *Geografiska Annaler. Series A, Physical Geography* 69 (1): 189–200. doi:10.1080/04353676.1987.11880207.

Walsh, K. C. 2007. "Identifying Suitable Sites for Wetland Mitigation at the Watershed Scale: A GIS Modeling Approach." The University of West Florida. http://etd.fcla.edu/WF/WFE0000100/Walsh_Kristal_Cobb_200712_MS.pdf .

Wen, L., X. Yang, and N. Saintilan. 2012. "Local Climate Determines the NDVI-Based Primary Productivity and Flooding Creates Heterogeneity in Semi-Arid Floodplain Ecosystem." *Ecological Modelling* 242: 116–126. doi:10.1016/j.ecolmodel.2012.05.018.

Weston, N. B. 2014. "Declining Sediments and Rising Seas: An Unfortunate Convergence for Tidal Wetlands." *Estuaries and Coasts* 37 (1): 1–23. doi:10.1007/s12237-013-9654-8.

Wu, W., M. Bethel, D. R. Mishra, and T. Hardy. 2018. "Model Selection in Bayesian Framework to Identify the Best WorldView-2 Based Vegetation Index in Predicting Green Biomass of Salt Marshes in the Northern Gulf of Mexico." *GIScience & Remote Sensing*. doi:10.1080/15481603.2018.1460934.

Wu, W., P. Biber, and M. Bethel. 2017. "Thresholds of Sea-Level Rise Rate and Sea-Level Rise Acceleration Rate in a Vulnerable Coastal Wetland." *Ecology and Evolution* 7 (24): 10890–10903. doi:10.1002/ece3.3550.

Wu, W., P. Biber, D. R. Mishra, and S. Ghosh. 2020. "Sea-Level Rise Thresholds for Stability of Salt Marshes in a Riverine versus a Marine Dominated Estuary." *Science of the Total Environment* 718: 137181. doi:10.1016/j.scitotenv.2020.137181.

Wu, W., P. D. Biber, M. S. Peterson, and C. Gong. 2012. "Modeling Photosynthesis of Spartina Alterniflora (Smooth Cordgrass) Impacted by the Deepwater Horizon Oil Spill Using Bayesian Inference." *Environmental Research Letters* 7 (4): 045302. doi:10.1088/1748-9326/7/4/045302.

Wu, W., J. S. Clark, and J. M. Vose. 2010. "Assimilating Multi-Source Uncertainties of a Parsimonious Conceptual Hydrological Model Using Hierarchical {bayesian} Modeling." *Journal of Hydrology* 394 (3–4): 436–446. doi:10.1016/j.jhydrol.2010.09.017.

Wu, W., K. M. Yeager, M. S. Peterson, and R. S. Fulford. 2015. "Neutral Models as a Way to Evaluate the Sea Level Affecting Marshes Model (SLAMM)." *Ecological Modelling* 303: 55–69. Elsevier B.V. doi:10.1016/j.ecolmodel.2015.02.008.

Yuill, B., D. Lavoie, and D. J. Reed. 2009. "Understanding Subsidence Processes in Coastal Louisiana." *Journal of Coastal Research* SI 54: 23–36. Special is. doi:10.2112/SI54-012.1.

Zar, J. H. 2010. *Biostatistical Analysis*. 5th ed. Upper Saddle River, New Jersey: Prentice Hall, Pearson Education Inc.

Zuur, A., E. N. Ieno, N. J. Walker, A. A. Saveliev, G. M. Smith, A. A. Saveliv, and G. M. Smith. 2009. *Mixed Effects Models and Extensions in Ecology with R*. New York: Springer.